



PT. SALIPI NAGARI PERKASA

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LAPORAN UJI MINYAK TRAFO

di

UNIT JASA PEMBANGKIT

JERANJANG



Yang dilakukan pada tanggal 14 NOVEMBER 2019

Untuk keadaan trafo periode 13 November ~ 08 December 2019

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1.0 Specific Information:

No.	Unit No.	Serial Number	TC	Test Requested
1.	MAIN TRAFO UNIT 1	100691	4	OLTT+ and Corrosive Sulfur
2.	TRAFO 1 ESP UNIT 1	10937	5	OLTT+ and Corrosive Sulfur
3.	TRAFO 2 ESP UNIT 1	10035	6	OLTT+ and Corrosive Sulfur
4.	TRAFO 3 ESP UNIT 1	10031	7	OLTT+ and Corrosive Sulfur
5.	TRAFO 4 ESP UNIT 1	10034	8	OLTT+ and Corrosive Sulfur
6.	MAIN TRAFO UNIT 3	3011160192	23	OLTT+ and Corrosive Sulfur
7.	UAT 3	PX18-0082	22	OLTT+ and Corrosive Sulfur
8.	TRAFO 1 ESP UNIT 3	0073084208	11	OLTT+ and Corrosive Sulfur
9.	TRAFO 2 ESP UNIT 3	0070084208	12	OLTT+ and Corrosive Sulfur
10.	TRAFO 3 ESP UNIT 3	0072084208	13	OLTT+ and Corrosive Sulfur
11.	TRAFO 4 UNIT 3	0071084208	14	OLTT+ and Corrosive Sulfur
12.	ASH HANDLING-1	10820108	15	OLTT+ and Corrosive Sulfur
13.	ASH HANDLING-2	10820107	16	OLTT+ and Corrosive Sulfur
14.	TRAFO 1 ESP UNIT 2	10037	17	OLTT+ and Corrosive Sulfur
15.	TRAFO 2 ESP UNIT 2	10036	18	OLTT+ and Corrosive Sulfur
16.	TRAFO 3 ESP UNIT 2	10033	19	OLTT+ and Corrosive Sulfur
17.	TRAFO 4 ESP UNIT 2	10032	20	OLTT+ and Corrosive Sulfur
18.	MAIN TRAFO UNIT 2	1006902	21	OLTT+ and Corrosive Sulfur

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Note:

- * OLTT+ consists of Liquid Screen Tests, Dissolved Gas Analysis (DGA), Water Content, Furan Analysis, Metals-in-Oil Analysis, Liquid Power Factors, and Inhibitor Content (DBPC).

- ** Liquid Screen Test consists of Acidity, Interfacial Tension (IFT), Dielectric, Colour, Specific Gravity, Visualization, and Sediment.

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2.0 Our Comments and Recommendations of The Test Results Analysis:

We had reviewed each laboratory test result against respective benchmark parameter and following are our comments and recommendations.

No.	Unit No.	Serial Number	TC
1.	MAIN TRAFO UNIT 1	100691	4

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 42.1mN/m are **ACCEPTABLE**.
- **Dielectric** of 74kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows combustible gasses levels to have stabilized. Proceed with annual monitoring.
- **Water Content** in oil of 6ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.43% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.23% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.014 AC		83 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.014 AC		73 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		75 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		71 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.017 AC	41.3 AC	66 AC	0.5 AC	0.8789 AC	CLEAR AC	
25/07/2019	ROUTINE	0.014 AC		79 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.014 AC		61 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.017 AC		73 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.017 AC		69 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.014 AC	42.1 AC	74 AC	0.5 AC	0.8713 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	147	9471	43278	32	655	2728	4	5	ND	843	56320
06/12/2017	183	12895	43794	39	715	3166	5	5	ND	947	60802
28/06/2018	176	11044	46776	41	758	3393	5	6	ND	986	62199
20/06/2019	18	19122	60738	26	756	3272	2	3	ND	805	83937
14/11/2019	21	21627	62797	15	652	3391	2	3	ND	693	88508

RECOMMENDATION:
GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	44	8	0.36	A
22/11/2018	41	9	0.45	A
08/12/2018	41	8	0.40	A
21/05/2019	38	8	0.52	A
20/06/2019	39	7	0.38	A
25/07/2019	38	8	0.52	A
20/08/2019	38	7	0.46	A
17/09/2019	35	6	0.43	A
17/10/2019	32	5	0.47	A
14/11/2019	35	6	0.43	A

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

4

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 100691
KVA	- 40000
PRI	- 165000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.1%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.02	0.01	0.01
06/12/2017	ND	0.01	0.04
28/06/2018	ND	ND	0.04
20/06/2019	0.01	ND	0.02
14/11/2019	ND	ND	0.03

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.140 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.140 AC
20/06/2019	0.000 AC	0.180 AC
14/11/2019	0.000 AC	0.140 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.24 AC
06/12/2017	0.27 AC
28/06/2018	0.24 AC
20/06/2019	0.24 AC
14/11/2019	0.23 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
2.	TRAFO 1 ESP UNIT 1	10937	5

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 38.4mN/m are **ACCEPTABLE**.
- **Dielectric** of 60kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 16ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.31% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected were less than 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

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DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	OFFLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.020 AC		49 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.017 AC		30 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.017 AC		41 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.017 AC		40 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.017 AC	38.7 AC	34 AC	0.5 AC	0.8747 AC	CLEAR AC	
25/07/2019	ROUTINE	0.017 AC		61 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.017 AC		56 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.020 AC		60 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.022 AC		54 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.020 AC	38.4 AC	60 AC	0.5 AC	0.8670 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	20494	67487	4	69	1143	ND	1	ND	109	89233
06/12/2017	39	19933	67198	4	83	1261	ND	1	ND	127	88519
28/06/2018	44	20039	64529	4	119	1696	ND	2	ND	169	86433
20/06/2019	22	27043	68613	5	36	689	ND	ND	ND	63	96408
14/11/2019	21	29985	68101	7	33	728	ND	ND	ND	61	98875

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	40	24	1.32	B
22/11/2018	37	23	1.50	C
08/12/2018	32	18	1.71	D
21/05/2019	39	22	1.21	B
20/06/2019	38	23	1.50	C
25/07/2019	29	18	1.89	D
20/08/2019	29	16	1.68	C
17/09/2019	28	13	1.53	C
17/10/2019	30	16	1.68	C
14/11/2019	33	16	1.31	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

5

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	95	ND	ND	ND	95
06/12/2017	ND	75	ND	ND	ND	75
28/06/2018	ND	85	ND	ND	ND	85
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10937
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.04

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.870 AC
06/12/2017	0.000 AC	0.570 AC
28/06/2018	0.000 AC	0.680 AC
20/06/2019	0.000 AC	0.230 AC
14/11/2019	0.000 AC	0.370 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	ND UN
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	1B	DARK ORANGE
14/11/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
3.	TRAFO 2 ESP UNIT 1	10035	6

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 32.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 61kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 14ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.15% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process in the oil.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

Page Disclaimer :

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						
25/07/2019	OFFLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.020 AC		63 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.020 AC		51 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.020 AC		71 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.020 AC	27.9 UN	68 AC	0.5 AC	0.8658 AC	CLEAR AC	
25/07/2019	ROUTINE	0.022 AC		67 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.020 AC		51 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.022 AC		47 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.020 AC		58 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.017 AC	32.3 AC	61 AC	0.5 AC	0.8583 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	27	18639	66781	3	64	1106	ND	1	ND	95	86621
06/12/2017	20	19360	61590	3	45	728	ND	1	ND	69	81747
28/06/2018	22	22494	64443	4	57	789	ND	ND	ND	83	87809
20/06/2019	23	17710	61707	5	23	842	ND	ND	ND	51	80310
14/11/2019	21	25310	59578	6	33	956	ND	ND	ND	60	85904

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	33	14	1.15	B
22/11/2018	34	18	1.48	C
08/12/2018	33	14	1.15	B
21/05/2019	37	15	0.98	B
20/06/2019	37	17	1.11	B
25/07/2019	30	15	1.58	C
20/08/2019	29	15	1.58	C
17/09/2019	28	16	1.89	D
17/10/2019	29	14	1.47	C
14/11/2019	33	14	1.15	B

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

6

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	62	ND	ND	ND	62
06/12/2017	ND	67	ND	ND	ND	67
28/06/2018	ND	84	ND	ND	ND	84
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10035
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.900 AC
06/12/2017	0.000 AC	0.980 AC
28/06/2018	0.000 AC	1.310 AC
20/06/2019	0.000 AC	0.680 AC
14/11/2019	0.000 AC	0.360 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.02 UN
06/12/2017	0.02 UN
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
4.	TRAFO 3 ESP UNIT 1	10031	7

A) Analysis

- **Acidity** of 0.062mgKOH/gm and **IFT** of 29.3mN/m are **QUESTIONABLE**.
Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 48kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.48% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit.
Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

7

DATE PRINTED 04/12/2019

ID# - 601669

CUST - PT IP UJP PLTU
JERANJANG

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.056 QU		44 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.056 QU		39 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.056 QU		39 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.051 QU		33 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.051 QU	29.6 QU	30 AC	1.5 AC	0.8689 AC	CLEAR AC	
25/07/2019	ROUTINE	0.056 QU		48 AC	1.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.051 QU		43 AC	1.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.056 QU		48 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.059 QU		52 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.062 QU	29.3 QU	48 AC	1.5 AC	0.8613 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	32	15273	65111	5	95	1002	ND	2	ND	134	81520
06/12/2017	7	14730	65738	3	42	617	ND	ND	ND	52	81137
28/06/2018	9	17958	66007	4	46	781	ND	1	ND	60	84806
20/06/2019	16	17771	65478	6	73	829	ND	1	ND	96	84174
14/11/2019	13	19116	69791	7	58	909	ND	1	ND	79	89895

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	36	20	1.42	C
22/11/2018	36	22	1.56	C
08/12/2018	35	20	1.42	C
21/05/2019	36	19	1.35	B
20/06/2019	36	19	1.35	B
25/07/2019	31	15	1.42	C
20/08/2019	32	18	1.71	D
17/09/2019	32	18	1.71	D
17/10/2019	37	19	1.24	B
14/11/2019	34	18	1.48	C

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

7

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	32	ND	ND	ND	32
06/12/2017	ND	14	ND	ND	ND	14
28/06/2018	ND	20	ND	ND	ND	20
20/06/2019	ND	57	ND	ND	ND	57
14/11/2019	ND	56	ND	ND	ND	56

RECOMMENDATION:

FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	0.01	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.11
20/06/2019	0.01	ND	0.02
14/11/2019	ND	ND	0.06

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10031
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.700 AC
06/12/2017	0.000 AC	0.630 AC
28/06/2018	0.000 AC	1.030 AC
20/06/2019	0.000 AC	0.520 AC
14/11/2019	0.000 AC	0.550 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.01 UN
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1A	LIGHT ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
5.	TRAFO 4 ESP UNIT 1	10034	8

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 38.8mN/m are **ACCEPTABLE**.
- **Dielectric** of 58kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 17ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.39% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factor** for 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

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DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.014 AC		52 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.014 AC		44 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		47 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		41 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.014 AC	38.6 AC	45 AC	0.5 AC	0.8775 AC	CLEAR AC	
25/07/2019	ROUTINE	0.014 AC		67 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.017 AC		45 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.017 AC		52 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.020 AC		56 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.020 AC	38.8 AC	58 AC	0.5 AC	0.8699 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	13	12265	66770	5	172	1206	ND	8	ND	198	80439
06/12/2017	8	13331	63527	3	70	720	ND	1	ND	82	77660
28/06/2018	9	17962	64248	4	75	1054	ND	2	ND	90	83354
20/06/2019	11	15056	64828	5	11	951	ND	ND	ND	27	80862
14/11/2019	19	19904	66458	6	17	1109	ND	ND	ND	42	87513

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	41	21	1.05	B
22/11/2018	38	21	1.37	C
08/12/2018	38	20	1.30	B
21/05/2019	36	19	1.35	B
20/06/2019	37	19	1.24	B
25/07/2019	32	16	1.52	C
20/08/2019	33	16	1.31	B
17/09/2019	33	16	1.31	B
17/10/2019	32	18	1.71	D
14/11/2019	34	17	1.39	C

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

8

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 1**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10034
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	18	ND	ND	ND	18
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	21	ND	ND	ND	21
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.05	0.02	0.01
06/12/2017	ND	ND	0.02
28/06/2018	ND	ND	0.05
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.180 QU	6.680 UN
06/12/2017	0.010 AC	2.180 AC
28/06/2018	0.060 AC	2.850 AC
20/06/2019	0.010 AC	1.120 AC
14/11/2019	0.010 AC	1.090 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.08 UN
06/12/2017	0.06 UN
28/06/2018	0.02 UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE SERVICE
01/08/2018 CHANGED OIL

No.	Unit No.	Serial Number	TC
6.	MAIN TRAF0 UNIT 3	3011160192	23

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 39.4mN/m are **ACCEPTABLE**.
- **Dielectric** of 69kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 7ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.38% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

23

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- PAUWES TRAFU
DATE MFG	- 2018
MAIN TANK S/N	- 3011160192
KVA	- 60000
PRI	- 157000 V
SEC	- 6300 V
KILOGRAMS	- 103500
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.171%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30183

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2019	ROUTINE	0.011 AC		93 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.011 AC	40.6 AC	83 AC	0.5 AC	0.8786 AC	CLEAR AC	
25/07/2019	ROUTINE	0.014 AC		87 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.011 AC		78 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.014 AC		81 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.017 AC		85 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.014 AC	39.4 AC	69 AC	0.5 AC	0.8709 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
20/06/2019	3	9903	31346	6	156	369	ND	ND	ND	165	41783
14/11/2019	13	9649	34449	8	170	610	ND	ND	ND	191	44899

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2019	38	7	0.46	A
20/06/2019	38	8	0.52	A
25/07/2019	37	6	0.39	A
20/08/2019	39	6	0.33	A
17/09/2019	38	6	0.39	A
17/10/2019	43	7	0.31	A
14/11/2019	39	7	0.38	A

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

23

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- PAUWES TRAFU
DATE MFG	- 2018
MAIN TANK S/N	- 3011160192
KVA	- 60000
PRI	- 157000 V
SEC	- 6300 V
KILOGRAMS	- 103500
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAF
IMPEDENCE	- 13.171%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30183
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
20/06/2019	ND	ND	0.02
14/11/2019	0.01	ND	0.05

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
20/06/2019	0.000 AC	0.090 AC
14/11/2019	0.000 AC	0.410 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
7.	UAT 3	PX18-0082	22

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 35.1mN/m are **ACCEPTABLE**.
- **Dielectric** of 85kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 6ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.39% M/dw is graded as **A**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes.**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

22

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- B & D
DATE MFG	- 2018
MAIN TANK S/N	- PX18-0082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 52000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 12000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
21/05/2019	ROUTINE	0.014 AC		85 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.014 AC	37.0 AC	89 AC	0.5 AC	0.8798 AC	CLEAR AC	
25/07/2019	ROUTINE	0.014 AC		83 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.014 AC		78 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.014 AC		88 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.014 AC		90 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.014 AC	35.1 AC	85 AC	0.5 AC	0.8721 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
20/06/2019	4	10251	34223	16	39	232	26	2	ND	87	44793
14/11/2019	5	11577	37124	16	59	338	28	2	ND	110	49149

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
21/05/2019	35	5	0.35	A
20/06/2019	34	5	0.41	A
25/07/2019	32	5	0.47	A
20/08/2019	35	4	0.28	A
17/09/2019	36	4	0.28	A
17/10/2019	37	3	0.19	A
14/11/2019	37	6	0.39	A

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

22

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **UAT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- B & D
DATE MFG	- 2018
MAIN TANK S/N	- PX18-0082
KVA	- 6300
PRI	- 6300 V
SEC	- 6300 V
KILOGRAMS	- 52000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.78%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 12000
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	10	ND	ND	ND	10

RECOMMENDATION:
FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A
HEALTHY SOLID INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.05

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
20/06/2019	0.000 AC	0.120 AC
14/11/2019	0.000 AC	0.190 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
8.	TRAFO 1 ESP UNIT 3	0073084208	11

A) Analysis

- **Acidity** of 0.020mgKOH/g and **IFT** of 35.7mN/m are **ACCEPTABLE**.
- **Dielectric** of 78kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 8ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.84% M/dw is graded as **A**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.16% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

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DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0073084208
KVA	-
PRI	- 72000 V
SEC	- 416 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.020 AC		63 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.020 AC		53 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.020 AC		62 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		85 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	35.2 AC	65 AC	0.5 AC	0.8212 QU	CLEAR AC	
25/07/2019	ROUTINE	0.020 AC		51 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.020 AC		69 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.022 AC		67 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.022 AC		76 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.020 AC	35.7 AC	78 AC	0.5 AC	0.8140 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	14	12943	68807	3	62	1243	ND	ND	ND	79	83072
06/12/2017	14	12739	65097	3	66	612	ND	ND	ND	83	78531
28/06/2018	19	15552	64448	3	57	1391	ND	ND	ND	79	81470
20/06/2019	11	19448	64974	6	15	1064	ND	ND	ND	32	85518
14/11/2019	16	20510	67519	6	31	1063	ND	ND	ND	53	89145

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	31	12	1.14	B
22/11/2018	32	9	0.86	B
08/12/2018	29	9	0.94	B
21/05/2019	25	9	1.17	B
20/06/2019	26	9	1.17	B
25/07/2019	25	9	1.17	B
20/08/2019	26	8	1.04	B
17/09/2019	26	8	1.04	B
17/10/2019	29	8	0.84	A
14/11/2019	29	8	0.84	A

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

11

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG - ZHEJIAN
DATE MFG - 2010
MAIN TANK S/N - **0073084208**
KVA -
PRI - 72000 V
SEC - 416 V
KILOGRAMS - 1350
EQUIP TYPE - TRANSFORMER
TRANS CLASS - ONAN
IMPEDENCE -
PHASE/CYCLE - 3/50 Hz
LIQUID TYPE - OIL
LITRES LIQUID - 731

LTC MFG -
LTC S/N -
LTC FAB NO -
LTC MODEL -
LTC LITRES LIQUID -
LTC ENERGIZED -

ADDITIONAL EQUIPMENT

RADIATORS - YES
FANS - NO
H2O COOLED - NO
OIL PUMPS - NO
TOP FPV -
BOTTOM FPV -
BUCHOLZ - NO
CONSERVATOR - YES
LTC COMPARTMENT - NO
BUSHING LOC - TOP
FREE BREATHER - YES
HOSE LENGTH -
POWER AVAILABLE - YES

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	0.01	0.01	ND
06/12/2017	0.01	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.350 AC
06/12/2017	0.000 AC	0.040 AC
28/06/2018	0.000 AC	0.050 AC
20/06/2019	0.000 AC	0.060 AC
14/11/2019	0.000 AC	0.240 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.18 QU
28/06/2018	0.16 QU
20/06/2019	0.16 QU
14/11/2019	0.16 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE

No.	Unit No.	Serial Number	TC
9.	TRAFO 2 ESP UNIT 3	0070084208	12

A) Analysis

- **Acidity** of 0.028mgKOH/g is **ACCEPTABLE** while **IFT** of 31.1mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 48kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water Content** in oil of 24ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 1.56% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 20ppm. This indicates higher than normal moisture content in the oil.
- **Furanic compound** 2FAL of 161ppb was detected which indicates normal aging of the solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

Page Disclaimer :

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B) Recommendation

- Monitor the **Specific Gravity** of the oil every 12 months.
- Retest **Moisture** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

12

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.025 AC		54 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.025 AC		33 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.025 AC		30 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.028 AC		40 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.028 AC	31.3 QU	41 AC	1.0 AC	0.8163 QU	CLEAR AC	
25/07/2019	ROUTINE	0.031 AC		60 AC	1.0 AC		CLEAR AC	
20/08/2019	ROUTINE	0.028 AC		48 AC	1.0 AC		CLEAR AC	
17/09/2019	ROUTINE	0.031 AC		43 AC	1.0 AC		CLEAR AC	
17/10/2019	ROUTINE	0.031 AC		45 AC	1.0 AC		CLEAR AC	
14/11/2019	ROUTINE	0.028 AC	31.1 QU	48 AC	1.0 AC	0.8093 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	38	16781	70523	5	129	2431	3	5	ND	180	89915
06/12/2017	42	19523	66059	3	102	912	ND	ND	ND	147	86641
28/06/2018	43	22272	68719	4	38	1226	ND	ND	ND	85	92302
20/06/2019	13	21221	65168	7	48	1870	ND	ND	ND	68	88327
14/11/2019	11	23917	69704	6	48	1925	ND	ND	ND	65	95611

RECOMMENDATION:

RETEST 12 MONTHS:

TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	32	13	1.24	B
22/11/2018	42	38	1.90	D
08/12/2018	43	32	1.44	C
21/05/2019	33	18	1.48	C
20/06/2019	32	18	1.71	D
25/07/2019	29	21	2.20	D
20/08/2019	31	19	1.80	D
17/09/2019	31	20	1.90	D
17/10/2019	34	19	1.56	C
14/11/2019	37	24	1.56	C

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

12

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	169	ND	ND	169
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	180	ND	ND	180
14/11/2019	ND	ND	161	ND	ND	161

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0070084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.01

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.330 AC
06/12/2017	0.000 AC	0.030 AC
28/06/2018	0.000 AC	0.040 AC
20/06/2019	0.000 AC	0.010 AC
14/11/2019	0.000 AC	0.010 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	ND UN
06/12/2017	0.20 AC
28/06/2018	0.19 QU
20/06/2019	0.19 QU
14/11/2019	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	3A	MAGENTA
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
10.	TRAFO 3 ESP UNIT 3	0072084208	13

A) Analysis

- **Acidity** of 0.017mgKOH/g and **IFT** of 37.0mN/m are **ACCEPTABLE**.
- **Dielectric** of 34kV is **ACCEPTABLE**.
- **Colour** and **Visualization** for the oil are within the **ACCEPTABLE** range.
- **Specific Gravity** value is inside the **QUESTIONABLE** range. Close monitoring is needed and the cause should be identified.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 26ppm is **UNACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.17% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm. The presence of high water content will act as catalyst to the oxidation process and could cause dielectric failure.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.18% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur** test on copper strip resultant in colour changes to silvery, thus **Non-corrosive of 2D**.

Page Disclaimer :

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B) Recommendation

- We recommend that this transformer's oil be subjected **OIL PURIFICATION**.
Result indicates that the water content in the transformer oil is not conducive for normal operation of a transformer. It is advisable to remove the moisture in the transformer oil to improve dielectric breakdown voltage value. Furthermore, oil inhibitor 2,6-DBPC and passivator could be added during oil purification.
- Alternatively, oil change can be performed in lieu of purification to solve the bad Specific Gravity values problem. Kindly consult the manufacture on any suitable oil to replace the existing oil-filled.

Page Disclaimer :

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

13

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.014 AC		49 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.017 AC		39 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.014 AC		34 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.014 AC		41 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.020 AC	38.5 AC	36 AC	0.5 AC	0.8215 QU	CLEAR AC	
25/07/2019	ROUTINE	0.020 AC		36 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.017 AC		32 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.020 AC		43 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.017 AC		36 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.017 AC	37.0 AC	34 AC	0.5 AC	0.8143 QU	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	21	16364	70694	3	38	1068	ND	ND	ND	62	88188
06/12/2017	15	14515	65277	2	40	559	ND	ND	ND	57	80408
28/06/2018	30	15202	64188	4	65	1833	ND	ND	ND	99	81322
20/06/2019	14	18302	64811	6	86	2181	ND	ND	ND	106	85400
14/11/2019	10	16999	64949	8	91	1921	ND	ND	ND	109	83978

RECOMMENDATION:

RETEST 12 MONTHS:

TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	39	15	0.82	A
22/11/2018	40	15	0.82	A
08/12/2018	43	18	0.81	A
21/05/2019	36	20	1.42	C
20/06/2019	35	23	1.63	C
25/07/2019	31	20	1.90	D
20/08/2019	33	19	1.56	C
17/09/2019	33	20	1.64	C
17/10/2019	42	21	1.05	B
14/11/2019	43	26	1.17	B

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

13

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	4	ND	ND	ND	4
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0072084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	ND	ND
28/06/2018	ND	ND	0.02
20/06/2019	0.01	ND	0.02
14/11/2019	ND	ND	0.02

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.620 AC
06/12/2017	0.000 AC	0.080 AC
28/06/2018	0.000 AC	0.010 AC
20/06/2019	0.000 AC	0.070 AC
14/11/2019	0.000 AC	0.110 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.21 AC
28/06/2018	0.20 AC
20/06/2019	0.19 QU
14/11/2019	0.18 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	2C	LAVENDER BLUE
06/12/2017	NON-COR	2E	BRASSY GOLD
28/06/2018	NON-COR	2E	BRASSY GOLD
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	2D	SILVERY

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
11.	TRAFO 4 UNIT 3	0071084208	14

A) Analysis

- **Acidity** of 0.025mgKOH/g is **ACCEPTABLE** while **IFT** of 31.3mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 49kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.26% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

Page Disclaimer :

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

14

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
25/09/2018	ONLINE						
23/10/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	OFFLINE						
17/09/2019	OFFLINE						
17/10/2019	OFFLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.022 AC		56 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		44 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		49 AC	1.0 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		58 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	30.3 QU	51 AC	1.0 AC	0.8451 AC	CLEAR AC	
25/07/2019	ROUTINE	0.020 AC		75 AC	1.0 AC		CLEAR AC	
20/08/2019	ROUTINE	0.022 AC		49 AC	1.0 AC		CLEAR AC	
17/09/2019	ROUTINE	0.025 AC		51 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.028 AC		51 AC	1.0 AC		CLEAR AC	
14/11/2019	ROUTINE	0.025 AC	31.3 QU	49 AC	1.0 AC	0.8440 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	10	18669	66036	3	24	665	ND	ND	ND	37	85407
06/12/2017	14	15414	63712	2	22	675	ND	ND	ND	38	79839
28/06/2018	20	14690	67822	2	22	683	ND	ND	ND	44	83239
20/06/2019	19	16312	66417	4	37	732	ND	ND	ND	60	83521
14/11/2019	13	18447	65198	6	34	877	ND	ND	ND	53	84575

RECOMMENDATION:

RETEST 12 MONTHS:

TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	30	12	1.26	B
22/11/2018	30	11	1.15	B
08/12/2018	29	13	1.37	C
21/05/2019	26	11	1.43	C
20/06/2019	27	12	1.42	C
25/07/2019	26	11	1.43	C
20/08/2019	27	11	1.30	B
17/09/2019	27	12	1.42	C
17/10/2019	28	14	1.65	C
14/11/2019	29	12	1.26	B

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

14

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 UNIT 3**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

NAMEPLATE DATA

MFG	- ZHEJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 0071084208
KVA	-
PRI	- 72000 V
SEC	- 415 V
KILOGRAMS	- 1350
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 731

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	ND	ND
06/12/2017	ND	0.03	ND
28/06/2018	ND	0.03	0.02
20/06/2019	ND	0.03	0.01
14/11/2019	ND	ND	0.03

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.360 AC
06/12/2017	0.000 AC	0.370 AC
28/06/2018	0.000 AC	0.550 AC
20/06/2019	0.000 AC	0.490 AC
14/11/2019	0.000 AC	0.340 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.01 UN
06/12/2017	0.03 UN
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1B	DARK ORANGE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	1B	DARK ORANGE
14/11/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
12.	ASH HANDLING-1	10820108	15

A) Analysis

- **Acidity** of 0.028mgKOH/g and **IFT** of 38.2mN/m are **ACCEPTABLE**.
- **Dielectric** of 47kV is **ACCEPTABLE**.
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **DGA** shows high Carbon Monoxide and Carbon Dioxide gas level. However, the CO₂/CO ratio is within the tolerable range while other gas did not indicate any fault afflicting the transformer unit. For this case, may proceed with annual condition monitoring.
- **Water Content** in oil of 25ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 2.05% M/dw is graded as **C**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.21% by weight is detected, thus **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes**.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

15

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	ONLINE						
22/11/2018	ONLINE						
08/12/2018	ONLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		25 QU	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		20 UN	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		28 QU	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	36.6 AC	27 QU	1.5 AC	0.8890 AC	CLEAR AC	
25/07/2019	ROUTINE	0.025 AC		49 AC	1.0 AC		CLEAR AC	
20/08/2019	ROUTINE	0.022 AC		30 AC	1.0 AC		CLEAR AC	
17/09/2019	ROUTINE	0.025 AC		52 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.025 AC		57 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.028 AC	38.2 AC	47 AC	1.0 AC	0.8811 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	35	10984	71058	20	441	5037	7	8	ND	511	87590
06/12/2017	49	12034	67124	20	558	5440	8	8	ND	643	85241
28/06/2018	53	10939	69394	20	557	5425	9	8	ND	647	86405
20/06/2019	53	14032	65585	13	487	5360	ND	11	ND	564	85541
14/11/2019	42	13399	67123	22	661	6026	11	14	ND	750	87298

RECOMMENDATION:

CARBON MONOXIDE AND CARBON DIOXIDE LEVEL ARE HIGH. OTHER GASES LEVELS ARE ASSESSED AS ACCEPTABLE.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	36	29	2.06	C
22/11/2018	36	28	1.99	B
08/12/2018	35	28	1.99	B
21/05/2019	34	29	2.38	C
20/06/2019	34	29	2.38	C
25/07/2019	35	21	1.49	B
20/08/2019	38	20	1.30	B
17/09/2019	35	21	1.49	B
17/10/2019	36	22	1.56	B
14/11/2019	34	25	2.05	C

FIELD SERVICE

DATE	SERVICE
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CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

15

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-1**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	21	ND	ND	ND	21

RECOMMENDATION:

FURAN ANALYSIS SHOWS MINOR DETECTION OF FURANIC COMPOUNDS. THIS IS IN LINE WITH A HEALTHY SOLID INSULATION.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820108
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.56%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	®ALUMINIUM	®IRON	COPPER
04/07/2017	ND	0.01	0.01
06/12/2017	ND	ND	0.01
28/06/2018	ND	0.01	0.04
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	0.730 AC
06/12/2017	0.000 AC	0.240 AC
28/06/2018	0.000 AC	0.460 AC
20/06/2019	0.000 AC	0.360 AC
14/11/2019	0.000 AC	0.230 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.23 AC
06/12/2017	0.24 AC
28/06/2018	0.22 AC
20/06/2019	0.21 AC
14/11/2019	0.21 AC

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	1B	DARK ORANGE
06/12/2017	NON-COR	2C	LAVENDER BLUE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
13.	ASH HANDLING-2	10820107	16

A) Analysis

- **Acidity** of 0.022mgKOH/g and **IFT** of 34.6mN/m are **ACCEPTABLE**.
- **Dielectric** of 29kV is **QUESTIONABLE**. This is reflected from the high water content detected in the transformer oil
- **Specific gravity, Colour, and Visualization** of oil is in the **ACCEPTABLE** range.
- **Dissolved Gas Analysis** shows Carbon Monoxide and Carbon Dioxide gas levels to have stabilized. Other gasses are assessed as acceptable. Proceed with annual monitoring.
- **Water Content** in oil of 31ppm is **QUESTIONABLE**. Meanwhile Water Content in solid insulation of 2.54% M/dw is graded as **D**. For this transformer, it is recommended to maintain water content in oil at less than 30ppm. This indicates higher than normal moisture content in the oil.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.19% by weight is **QUESTIONABLE**. Insufficient oil inhibitor will cause the insulating oil to oxidize at a faster rate, thus degrading the paper insulation. Monitor the level annually.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

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B) Recommendation

- Retest **Moisture** and **Dielectric** in 6 months to assess if corrective work on the transformer oil is required.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Other parameters: Do carry out *Liquid Screen Tests, Water Content, Furan Analysis, Metals-In-Oil, Liquid Power Factor and Inhibitor Content tests **annually for Condition Monitoring purposes.**

Page Disclaimer :

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

16

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
25/08/2018	OFFLINE						
25/09/2018	OFFLINE						
08/12/2018	ONLINE						
21/05/2019	OFFLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	OFFLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.022 AC		36 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.022 AC		30 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.022 AC		32 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.022 AC		32 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.022 AC	36.8 AC	37 AC	1.5 AC	0.8889 AC	CLEAR AC	
25/07/2019	ROUTINE	0.025 AC		42 AC	1.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.025 AC		36 AC	1.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.028 AC		37 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.025 AC		37 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.022 AC	34.6 AC	29 QU	1.5 AC	0.8809 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
04/07/2017	18	21157	64224	4	129	1676	ND	1	ND	152	87209
06/12/2017	20	15791	63530	5	275	2783	ND	4	ND	304	82408
28/06/2018	32	22904	67139	4	249	3112	ND	5	ND	290	93445
20/06/2019	39	11205	66397	21	439	5998	11	9	ND	519	84119
14/11/2019	18	16725	64652	7	296	4058	ND	8	ND	329	85764

RECOMMENDATION:

GAS LEVELS HAVE REMAINED STABLE. PROCEED WITH ANNUAL MONITORING.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	37	23	1.50	B
22/11/2018	36	25	1.77	B
08/12/2018	37	24	1.56	B
21/05/2019	35	23	1.63	B
20/06/2019	34	24	1.97	B
25/07/2019	35	27	1.92	B
20/08/2019	38	29	1.88	B
17/09/2019	34	26	2.13	C
17/10/2019	33	27	2.21	C
14/11/2019	33	31	2.54	D

FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

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DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **ASH HANDLING-2**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
04/07/2017	ND	ND	ND	ND	ND	ND
06/12/2017	ND	ND	3	ND	ND	3
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 10820107
KVA	- 1600
PRI	- 6300 V
SEC	- 400 V
KILOGRAMS	- 4505
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	- 5.6%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 1023

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
04/07/2017	ND	0.01	ND
06/12/2017	ND	0.01	0.01
28/06/2018	ND	ND	0.02
20/06/2019	0.01	ND	0.02
14/11/2019	ND	ND	0.06

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- NO
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- NO
HOSE LENGTH	-
POWER AVAILABLE	- YES

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
04/07/2017	0.000 AC	1.350 AC
06/12/2017	0.000 AC	0.270 AC
28/06/2018	0.000 AC	0.490 AC
20/06/2019	0.000 AC	0.680 AC
14/11/2019	0.000 AC	0.550 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
04/07/2017	0.22 AC
06/12/2017	0.24 AC
28/06/2018	0.20 AC
20/06/2019	0.20 AC
14/11/2019	0.19 QU

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
04/07/2017	NON-COR	3A	MAGENTA
06/12/2017	NON-COR	1A	LIGHT ORANGE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE SERVICE
01/07/2018 PURIFIED

No.	Unit No.	Serial Number	TC
14.	TRAFO 1 ESP UNIT 2	10037	17

A) Analysis

- **Acidity** of 0.031mgKOH/gm is **ACCEPTABLE**, while **IFT** of 30.7mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 41kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 15ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.23% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

Page Disclaimer :

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	ONLINE						
17/09/2019	OFFLINE						
17/10/2019	ONLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.028 AC		54 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.028 AC		54 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.028 AC		46 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.031 AC		67 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.031 AC	31.0 QU	62 AC	1.0 AC	0.8776 AC	CLEAR AC	
25/07/2019	ROUTINE	0.031 AC		64 AC	1.0 AC		CLEAR AC	
20/08/2019	ROUTINE	0.034 AC		63 AC	1.0 AC		CLEAR AC	
17/09/2019	ROUTINE	0.036 AC		19 UN	1.0 AC		CLEAR AC	
17/10/2019	ROUTINE	0.037 AC		91 AC	1.0 AC		CLEAR AC	
14/11/2019	ROUTINE	0.031 AC	30.7 QU	41 AC	1.0 AC	0.8700 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	93	17225	43231	5	57	775	ND	2	ND	157	61388
20/06/2019	62	18560	46166	5	50	839	ND	ND	ND	117	65682
14/11/2019	17	18543	51095	6	35	992	ND	ND	ND	58	70688

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	31	11	1.04	B
22/11/2018	32	10	0.95	B
08/12/2018	30	10	1.05	B
21/05/2019	30	12	1.26	B
20/06/2019	30	13	1.37	C
25/07/2019	29	11	1.15	B
20/08/2019	29	11	1.15	B
17/09/2019	28	62	7.32	E
17/10/2019	34	16	1.31	B
14/11/2019	33	15	1.23	B

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

17

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 1 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10037
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	0.06	0.03
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.030 AC	0.420 AC
20/06/2019	0.000 AC	0.720 AC
14/11/2019	0.010 AC	0.610 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
15.	TRAFO 2 ESP UNIT 2	10036	18

A) Analysis

- **Acidity** of 0.062mgKOH/g and **IFT** of 30.5mN/m are **QUESTIONABLE**.
Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 77kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 12ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.98% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in colour changes to brassy and gold, thus **Non-corrosive of 2E**.

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

18

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	ONLINE						
17/09/2019	OFFLINE						
17/10/2019	ONLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.062 QU		67 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.062 QU		61 AC	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.062 QU		59 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.065 QU		77 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.065 QU	29.7 QU	76 AC	1.0 AC	0.8655 AC	CLEAR AC	
25/07/2019	ROUTINE	0.062 QU		47 AC	1.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.065 QU		41 AC	1.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.067 QU		59 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.065 QU		74 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.062 QU	30.5 QU	77 AC	1.0 AC	0.8586 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	8	22714	64906	5	67	879	ND	ND	ND	80	88579
20/06/2019	5	29439	65338	5	25	977	ND	ND	ND	35	95789
14/11/2019	5	28110	64911	6	43	1069	ND	ND	ND	54	94144

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	31	13	1.24	B
22/11/2018	33	8	0.66	A
08/12/2018	30	8	0.84	A
21/05/2019	30	11	1.15	B
20/06/2019	31	12	1.14	B
25/07/2019	28	11	1.30	B
20/08/2019	30	12	1.26	B
17/09/2019	27	11	1.30	B
17/10/2019	32	11	1.04	B
14/11/2019	33	12	0.98	B

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

18

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 2 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10036
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	42	ND	ND	ND	42
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.680 AC
20/06/2019	0.000 AC	0.450 AC
14/11/2019	0.000 AC	0.640 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	2E	BRASSY GOLD

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
16.	TRAFO 3 ESP UNIT 2	10033	19

A) Analysis

- **Acidity** of 0.062mgKOH/g is **QUESTIONABLE** while **IFT** of 26.5mN/m is **UNACCEPTABLE**. Although the level remained stable, the oil are containing oxidation by-products. These oxidation by-products are depositing and degrading the oil as well as the paper insulation. Therefore, removal of these contaminations are highly recommended.
- **Dielectric** of 71kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 13ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.07% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at 25°C and 100°C are **ACCEPTABLE**.
- No **inhibitor content** is detectable, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- **Corrosive sulfur** test on copper strip resultant in magenta, thus **Non-corrosive of 3A**.

Page Disclaimer :

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B) Recommendation

- It is recommended to perform **Transformer Oil Regeneration** on this unit. Transformer oil regeneration at this stage of contamination will effectively remove unwanted oxidation byproducts from the insulating oil. This process will restore the oil back to its original quality, thus protecting the paper insulation and extending the transformer serviceable lifespan. Furthermore, oil inhibitor 2,6-DBPC and passivator will be added to preserve both the transformer oil and copper surfaces to enhance the service ability of the transformer.

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

19

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	ONLINE						
17/09/2019	OFFLINE						
17/10/2019	ONLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.070 QU		86 AC	1.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.070 QU		13 UN	1.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.070 QU		8 UN	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.073 QU		50 AC	1.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.073 QU	26.5 UN	53 AC	1.5 AC	0.8687 AC	CLEAR AC	
25/07/2019	ROUTINE	0.070 QU		48 AC	1.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.067 QU		58 AC	1.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.062 QU		84 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.067 QU		77 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.062 QU	26.5 UN	71 AC	1.5 AC	0.8616 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	14	17627	48672	4	47	935	ND	ND	ND	65	67299
20/06/2019	16	17449	51923	5	58	1049	ND	ND	ND	79	70500
14/11/2019	18	21705	58102	6	59	1088	ND	ND	ND	83	80978

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	32	13	1.24	B
22/11/2018	33	84	6.89	E
08/12/2018	31	84	7.98	E
21/05/2019	30	13	1.37	C
20/06/2019	30	13	1.37	C
25/07/2019	29	13	1.37	C
20/08/2019	31	14	1.33	B
17/09/2019	30	10	1.05	B
17/10/2019	33	12	0.98	B
14/11/2019	33	13	1.07	B

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

19

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 3 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10033
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.02
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.04

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.570 AC
20/06/2019	0.010 AC	0.790 AC
14/11/2019	0.000 AC	0.580 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	ND UN
20/06/2019	ND UN
14/11/2019	ND UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	3A	MAGENTA
20/06/2019	NON-COR	3A	MAGENTA
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
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FIELD SERVICE

DATE	SERVICE
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No.	Unit No.	Serial Number	TC
17.	TRAFO 4 ESP UNIT 2	10032	20

A) Analysis

- **Acidity** of 0.034mgKOH/gm is **ACCEPTABLE**, while **IFT** of 30.2mN/m is **QUESTIONABLE**. The levels remained stable without further oil degradation. Stabilized condition may proceed with annual monitoring.
- **Dielectric** of 50kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 18ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 1.48% M/dw is graded as **C**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** shows minor detection of furanic compounds. This is in line with a healthy solid insulation.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Oil Inhibitor** of 0.04% by weight is detected, thus **UNACCEPTABLE**. Lack of inhibitor content will accelerate the oxidation process which produces acid and sludge. These oxidation by-products are depositing and degrading the oil as well as the paper insulation.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

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B) Recommendation

- It is recommended to add oil inhibitor (2,6-DBPC) until the acceptance level (IEEE Std 57-106 suggests 0.09% as minimum level) as soon as possible so as to retard oil oxidation process. By doing so, oil degradation will be slowed down, hence protecting and preserving the transformer insulating material.
- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

20

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
23/10/2018	OFFLINE						
22/11/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	OFFLINE						
20/06/2019	OFFLINE						
25/07/2019	OFFLINE						
20/08/2019	ONLINE						
17/09/2019	OFFLINE						
17/10/2019	ONLINE						
14/11/2019	OFFLINE						

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.034 AC		57 AC	1.0 AC		CLEAR AC	
22/11/2018	ROUTINE	0.034 AC		69 AC	1.0 AC		CLEAR AC	
08/12/2018	ROUTINE	0.034 AC		61 AC	1.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.036 AC		60 AC	1.0 AC		CLEAR AC	
20/06/2019	ROUTINE	0.036 AC	30.3 QU	61 AC	1.0 AC	0.8732 AC	CLEAR AC	
25/07/2019	ROUTINE	0.034 AC		59 AC	1.0 AC		CLEAR AC	
20/08/2019	ROUTINE	0.036 AC		45 AC	1.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.034 AC		50 AC	1.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.037 AC		78 AC	1.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.034 AC	30.2 QU	50 AC	1.0 AC	0.8659 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	22	25201	74464	5	85	823	ND	ND	ND	112	100600
20/06/2019	33	25208	70145	4	35	853	ND	ND	ND	72	96278
14/11/2019	16	24508	71482	5	31	793	ND	ND	ND	52	96835

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	32	15	1.42	C
22/11/2018	33	14	1.15	B
08/12/2018	31	18	1.71	D
21/05/2019	31	14	1.33	B
20/06/2019	30	14	1.47	C
25/07/2019	29	12	1.26	B
20/08/2019	32	12	1.14	B
17/09/2019	30	12	1.26	B
17/10/2019	34	12	0.98	B
14/11/2019	33	18	1.48	C

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

20

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **TRAFO 4 ESP UNIT 2**

LOCATION - OUTDOOR
OTHER -

NAMEPLATE DATA

MFG	- FUJIAN
DATE MFG	- 2010
MAIN TANK S/N	- 10032
KVA	-
PRI	- 72000 V
SEC	- 380 V
KILOGRAMS	- 1000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- ONAN
IMPEDENCE	-
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 630
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- NO
H2O COOLED	- NO
OIL PUMPS	- NO
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- NO
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- NO

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
28/06/2018	ND	ND	ND	ND	ND	ND
20/06/2019	ND	7	ND	ND	ND	7
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF TRANSFORMER.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	1.620 AC
20/06/2019	0.020 AC	1.150 AC
14/11/2019	0.000 AC	0.960 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.01 UN
20/06/2019	0.01 UN
14/11/2019	0.04 UN

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	2C	LAVENDER BLUE
20/06/2019	NON-COR	2C	LAVENDER BLUE
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
------	---------

No.	Unit No.	Serial Number	TC
18.	MAIN TRAFO UNIT 2	1006902	21

A) Analysis

- **Acidity** of 0.014mgKOH/g and **IFT** of 39.3mN/m are **ACCEPTABLE**.
- **Dielectric** of 57kV is **ACCEPTABLE**.
- **Colour, Specific Gravity, and Visualization** for the oil are within the **ACCEPTABLE** range.
- **DGA** indicates the transformer is operating satisfactorily, without significant amount of combustible gases being generated.
- **Water content** in oil of 11ppm is **ACCEPTABLE**. Meanwhile Water Content in solid insulation of 0.71% M/dw is graded as **B**. For this transformer, it is advisable to maintain the water content level below 20ppm.
- **Furan analysis** indicates the solid insulation is normal. **None** of furanic compound is detected.
- **Metals-In-Oil** detected are below 2ppm, thus **ACCEPTABLE**.
- **Liquid Power Factors** at both 25°C and 100°C are **ACCEPTABLE**.
- **Inhibitor content** detected at 0.22% by weight is **ACCEPTABLE**.
- Tested on **copper** strip resultant in magenta overcoat on brassy strip, thus **Non-corrosive of 3A**.

B) Recommendation

- Add passivator before the sulfur convert to be corrosive enough to attack copper. Passivator would coat the copper surface to retard or delay attack by corrosive sulfur.
- Do carry-out *Liquid Screen, Liquid Power Factor, Inhibitor, Water Content, Metals-In-Oil, Furan Analysis & DGA periodically for **Condition Monitoring purposes**

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CENNIX LABORATORY (M) SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

21

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 2**

LOCATION - OUTDOOR
OTHER -

VISUAL INSPECTION

DATE	STATUS	OIL TEMP °C	W. TEMP °C	OIL LEVEL	PAINT	S. GEL	LEAKS
25/09/2018	ONLINE						
23/10/2018	OFFLINE						
08/12/2018	OFFLINE						
21/05/2019	ONLINE						
20/06/2019	ONLINE						
25/07/2019	ONLINE						
20/08/2019	ONLINE						
17/09/2019	ONLINE						
17/10/2019	ONLINE						
14/11/2019	ONLINE						

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857
LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

LIQUID SCREEN TEST DATA

DATE	SERVICES	ASTM D974 ACID (mgKOH/gm)	ASTM D971 IFT (mN/m)	IEC 156 DIEL (kV)	ASTM D1500 COLOUR	ASTM D4052 SP. GR.	ASTM D1524 VISUAL	ASTM D1698 ® SEDIMENT
23/10/2018	ROUTINE	0.011 AC		64 AC	0.5 AC		CLEAR AC	
22/11/2018	ROUTINE	0.011 AC		61 AC	0.5 AC		CLEAR AC	
08/12/2018	ROUTINE	0.011 AC		57 AC	0.5 AC		CLEAR AC	
21/05/2019	ROUTINE	0.011 AC		63 AC	0.5 AC		CLEAR AC	
20/06/2019	ROUTINE	0.011 AC	39.3 AC	58 AC	0.5 AC	0.8745 AC	CLEAR AC	
25/07/2019	ROUTINE	0.011 AC		49 AC	0.5 AC		CLEAR AC	
20/08/2019	ROUTINE	0.014 AC		36 AC	0.5 AC		CLEAR AC	
17/09/2019	ROUTINE	0.014 AC		63 AC	0.5 AC		CLEAR AC	
17/10/2019	ROUTINE	0.014 AC		65 AC	0.5 AC		CLEAR AC	
14/11/2019	ROUTINE	0.014 AC	39.3 AC	57 AC	0.5 AC	0.8671 AC	CLEAR AC	

DISSOLVED GAS ANALYSIS (ASTM D3612C), EXPRESSED IN PPM

DATE	HYDROGEN	OXYGEN	NITROGEN	METHANE	CARBON MONOXIDE	CARBON DIOXIDE	ETHANE	ETHYLENE	ACETYLENE	TOTAL COMBUSTIBLE	TOTAL GAS
28/06/2018	19	12733	53386	10	193	2448	ND	2	ND	224	68791
20/06/2019	10	17382	50681	8	107	712	ND	2	ND	127	68902
14/11/2019	10	13692	47807	7	72	736	ND	2	ND	91	62326

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT INDICATES NORMAL OPERATION.

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

DATE	TEMP°C	PPM	%M/dw	GRADE
23/10/2018	29	7	0.74	A
22/11/2018	32	7	0.67	A
08/12/2018	30	6	0.63	A
21/05/2019	35	7	0.50	A
20/06/2019	35	11	0.78	A
25/07/2019	32	7	0.67	A
20/08/2019	34	8	0.66	A
17/09/2019	34	8	0.66	A
17/10/2019	37	11	0.71	A
14/11/2019	38	11	0.71	A

FIELD SERVICE

DATE SERVICE

CENNIX LABORATORY SDN BHD

(Formerly known as Myers-Musteq Sdn Bhd)

21

DATE PRINTED 04/12/2019

ID# - 601669

**CUST - PT IP UJP PLTU
JERANJANG**

SUB NAME -
CITY - MATARAM LOMBOK
UNIT# - **MAIN TRAFU UNIT 2**

LOCATION - OUTDOOR
OTHER -

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

DATE	5H2F	2FOL	2FAL	2ACF	5M2F	TOTAL
06/12/2017	ND	ND	ND	ND	ND	ND
28/06/2018	ND	ND	ND	ND	113	113
20/06/2019	ND	ND	ND	ND	ND	ND
14/11/2019	ND	ND	ND	ND	ND	ND

RECOMMENDATION:
RETEST 12 MONTHS:
TEST RESULT OBTAINED IN THIS TEST IS IN LINE WITH A HEALTHY PAPER INSULATION OF
TRANSFORMER.

NAMEPLATE DATA

MFG	- WOLONG
DATE MFG	- 2011
MAIN TANK S/N	- 1006902
KVA	- 33000
PRI	- 170000 V
SEC	- 6300 V
KILOGRAMS	- 81000
EQUIP TYPE	- TRANSFORMER
TRANS CLASS	- OFAF
IMPEDENCE	- 13.03%
PHASE/CYCLE	- 3/50 Hz
LIQUID TYPE	- OIL
LITRES LIQUID	- 30857

LTC MFG	-
LTC S/N	-
LTC FAB NO	-
LTC MODEL	-
LTC LITRES LIQUID	-
LTC ENERGIZED	-

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

DATE	ALUMINIUM	IRON	COPPER
28/06/2018	ND	ND	0.03
20/06/2019	ND	ND	0.02
14/11/2019	ND	ND	0.03

ADDITIONAL EQUIPMENT

RADIATORS	- YES
FANS	- YES
H2O COOLED	- NO
OIL PUMPS	- YES
TOP FPV	-
BOTTOM FPV	-
BUCHOLZ	- YES
CONSERVATOR	- YES
LTC COMPARTMENT	- YES
BUSHING LOC	- TOP
FREE BREATHER	- YES
HOSE LENGTH	-
POWER AVAILABLE	- YES

LIQUID POWER FACTOR (ASTM D924)

DATE	25°C	100°C
28/06/2018	0.000 AC	0.410 AC
20/06/2019	0.010 AC	0.730 AC
14/11/2019	0.020 AC	0.600 AC

INHIBITOR CONTENT, DBPC (ASTM D2668)

DATE	%BY WEIGHT
28/06/2018	0.22 AC
20/06/2019	0.22 AC
14/11/2019	0.22 AC

CORROSIVE SULFUR (ASTM D1275)

DATE	STATUS	CLASS	APPEARANCE
28/06/2018	NON-COR	1B	DARK ORANGE
20/06/2019	NON-COR	2E	BRASSY GOLD
14/11/2019	NON-COR	3A	MAGENTA

PASSIVATOR COUNT (IEC 60666 - ANNEX B), EXPRESSED IN PPM

DATE	PPM
------	-----

FIELD SERVICE

DATE	SERVICE
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3.0 Summary of Our Comments & Recommendations:

No.	Sub Name	Unit No.	TC	Recommendation
1.	MAIN TRAFO UNIT 1	100691	4	- Add passivator - Periodic condition monitoring
2.	TRAFO 1 ESP UNIT 1	10937	5	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
3.	TRAFO 2 ESP UNIT 1	10035	6	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
4.	TRAFO 3 ESP UNIT 1	100 31	7	Perform OIL REGENERATION
5.	TRAFO 4 ESP UNIT 1	10034	8	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
6.	MAIN TRAFO UNIT 3	3011160192	23	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
7.	UAT 3	PX18-0082	22	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
8.	TRAFO 1 ESP UNIT 3	0073084208	11	- Retest 12 months: Specific Gravity - Add passivator - Other parameters: Periodic condition monitoring
9.	TRAFO 2 ESP UNIT 3	0070084208	12	- Retest 12 months: Specific Gravity - Retest 6 months: Dielectric and Water Content - Add passivator - Other parameters: Periodic condition monitoring

Page Disclaimer :

All information provided is deemed reliable but is not guaranteed. The interpretation, comments and recommendations are derived based on International Standards and are the personal opinion of the Analyst of TxM Services Sdn. Bhd. The customer is advised to independently verify the information herein

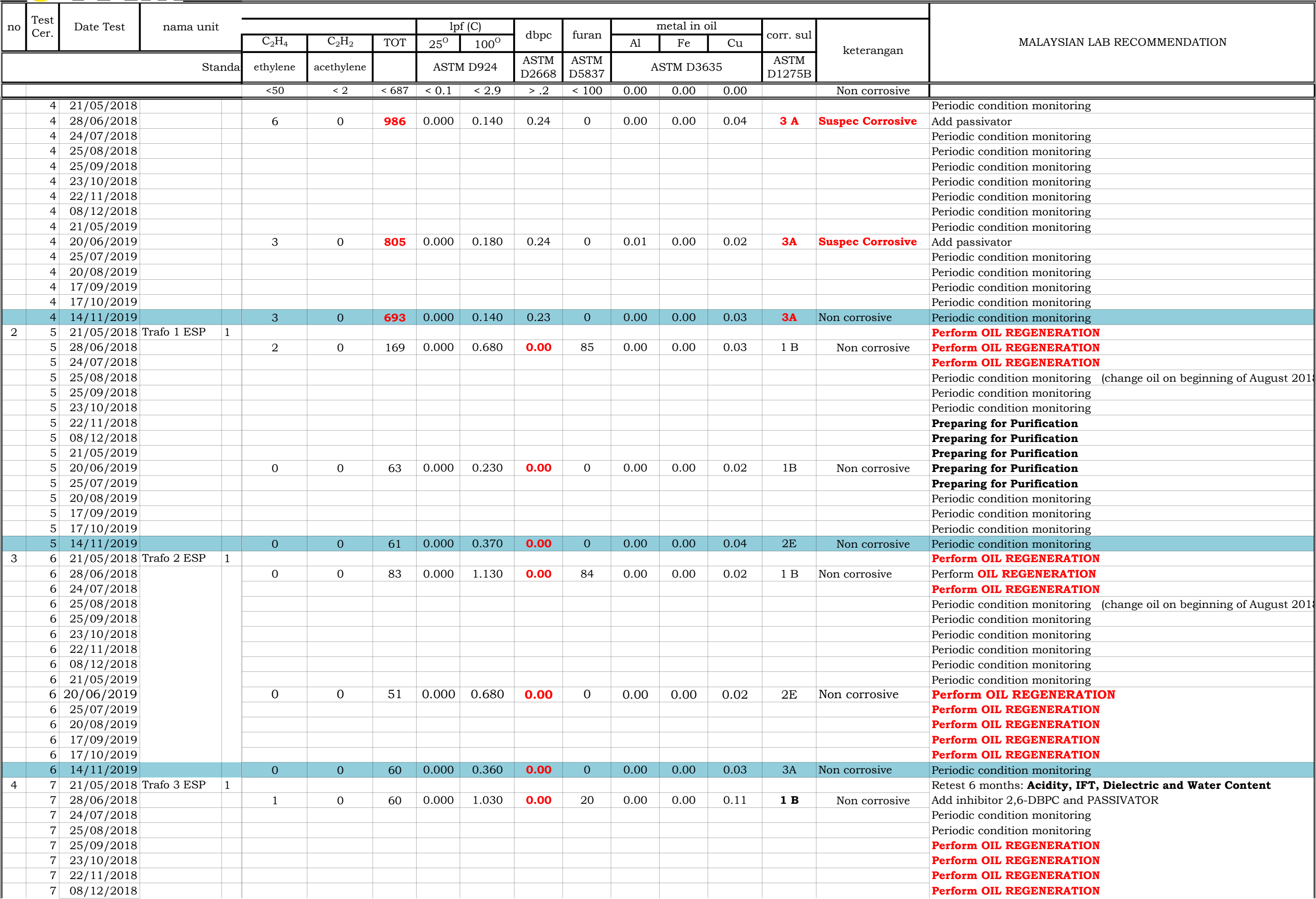
No.	Sub Name	Unit No.	TC	Recommendation
10.	TRAFO 3 ESP UNIT 3	0072084208	13	Perform OIL PURIFICATION
11.	TRAFO 4 UNIT 3	0071084208	14	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
12.	ASH HANDLING-1	10820108	15	- Add passivator - Periodic condition monitoring
13.	ASH HANDLING-2	10820107	16	- Retest 6 months: Dielectric and Water Content - Add passivator - Other parameters: Periodic condition monitoring
14.	TRAFO 1 ESP UNIT 2	10037	17	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
15.	TRAFO 2 ESP UNIT 2	10036	18	Perform OIL REGENERATION
16.	TRAFO 3 ESP UNIT 2	10033	19	Perform OIL REGENERATION
17.	TRAFO 4 ESP UNIT 2	10032	20	- Add inhibitor 2,6-DBPC and passivator - Periodic condition monitoring
18.	MAIN TRAFO UNIT 2	1006902	21	- Add passivator - Periodic condition monitoring

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no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	C ₂ H ₆
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	ethane	
			BATASAN OPERASI					< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	< 65
	4	21/05/2018						0.011		77	0.5		C		10	B							
	4	28/06/2018						0.014	41.9	69	0.5	0.8716	C		7	A	176	11,044	46,776	41	758	3,393	5
	4	24/07/2018						0.014		86	0.5		C		6	A							
	4	25/08/2018						0.017		90	0.5		C		6	A							
	4	25/09/2018						0.014		96	0.5		C		7	A							
	4	23/10/2018						0.014		83	0.5		C		8	A							
	4	22/11/2018						0.014		73	0.5		C		9	A							
	4	08/12/2018						0.014		75	0.5		C		8	A							
	4	21/05/2019						0.014		71	0.5		C		8	A							
	4	20/06/2019						0.017	41.3	66	0.5	0.8789	C		7	A	18	19,122	60,738	26	756	3,272	2
	4	25/07/2019						0.014		79	0.5		C		8	A							
	4	20/08/2019						0.014		61	0.5		C		7	A							
	4	17/09/2019						0.017		73	0.5		C		6	A							
	4	17/10/2019						0.017		69	0.5		C		5	A							
	4	14/11/2019						0.014	42.1	74	0.5	0.8713	C		6	A	21	21,627	62,797	15	652	3,391	2
2	5	21/05/2018	Trafo 1 ESP	1	10937	2010	1,143	Fujia	0.065		38	1.5	C		33	E							
	5	28/06/2018						0.067	25.9	32	1.5	0.8560	C		30	E	44	20,039	64,529	4	119	1,696	0
	5	24/07/2018						0.073		38	1.5		C		19	D							
	5	25/08/2018						0.020		71	0.5		C		12	B							
	5	25/09/2018						0.017		63	0.5		C		18	B							
	5	23/10/2018						0.020		49	0.5		C		24	B							
	5	22/11/2018						0.017		30	0.5		C		23	C							
	5	08/12/2018						0.017		41	0.5		C		18	D							
	5	21/05/2019						0.017		40	0.5		C		22	C							
	5	20/06/2019						0.017	38.7	34	0.5	0.8747	C		23	C	22	27,043	68,613	5		689	0
	5	25/07/2019						0.017		61	0.5		C		18	D							
	5	20/08/2019						0.017		56	0.5		C		16	C							
	5	17/09/2019						0.020		60	0.5		C		13	C							
	5	17/10/2019						0.022		54	0.5		C		16	C							
	5	14/11/2019						0.020	38.4	60	0.5	0.8670	C		16	B	21	29,985	68,101	7	33	728	0
3	6	21/05/2018	Trafo 2 ESP	1	10035	2010	1,143	Fujia	0.056		45	1.5	C		20	D							
	6	28/06/2018						0.059	27.6	55	1.5	0.8570	C		17	C	22	22,494	64,443	4	57	789	0
	6	24/07/2018						0.062		47	1.5		C		15	C							
	6	25/08/2018						0.022		56	0.5		C		13	B							
	6	25/09/2018						0.020		67	0.5		C		11	B							
	6	23/10/2018						0.020		71	0.5		C		14	B							
	6	22/11/2018						0.020		63	0.5		C		18	C							
	6	08/12/2018						0.020		51	0.5		C		14	B							
	6	21/05/2019						0.020		71	0.5		C		15	B							
	6	20/06/2019						0.020	27.9	68	0.5	0.8658	C		17	B	23	17,710	61,707	5	23	842	0
	6	25/07/2019						0.022		67	0.5		C		15	C							
	6	20/08/2019						0.020		51	0.5		C		15	C							
	6	17/09/2019						0.022		47	0.5		C		16	D							
	6	17/10/2019						0.020		58	0.5		C		14	C							
	6	14/11/2019						0.017	32.3	61	0.5	0.8583	C		14	B	21	25,310	59,578	6	33	956	0
4	7	21/05/2018	Trafo 3 ESP	1	10031	2010	1,143	Fujia	0.056		48	1.5	C		21	D							
	7	28/06/2018						0.053	29.7	62	1.5	0.8605	C		17	D	9	17,958	66,007	4	46	781	0
	7	24/07/2018						0.510		64	1.5		C		17	D							
	7	25/08/2018						0.056		57	1.5		C		15	C							
	7	25/09/2018						0.051		46	1.5		C		21	D							
	7	23/10/2018						0.056		44	1.5		C		20	C							
	7	22/11/2018						0.056		39	1.5		C		22	C							
	7	08/12/2018						0.056		39	1.5		C		20	C							





no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	4	21/05/2018		For Test 14 November 2019 ada beberapa perubahan kondisi dibandingkan bulan lalu. Kami meyarankan untuk penggantian Minyak untuk Trafo ESP 3 Unit 1 dan ESP 2 dan 3 Unit 2 PURIFIKASI dengan temperatur sekitar 70⁰ untuk ESP 2 & 3 Unit 3 serta kedua Trafo Ash Handling Khusus untuk kedua trafo Ash Handling , berdasarkan hasil DGA, perlu dilakukan penambahan <i>cooling system</i> atau pembersihan trafo agar <i>cooling system</i> lebih baik. For Test 17 October 2019 Kondidi masih sama seperti bulan sebelumnya, Kami tetap meyarankan untuk ESP 2 dan 3 Unit 3 dilakukan PURIFIKASI dengan temperatur sekitar 70⁰ GANTI MINYAK pada trafo ESP 2 dan 3 Unit 1 dan ESP 2 dan 3 Unit 2 Untuk Trafo ESP 1 Unit 2 yang mengalami perubahan mendadak, kemungkinannya adalah kondisi pelaksanaan sampling yang kurang baik pada bulan yang lalu. For Test 17 September 2019 Pada dasarnya masih sama seperti bulan sebelumnya, KECUALI trafo ESP 1 Unit 2 , terjadi kenaikan Water Content dan penurunan BDV yang sangat mencolok, Perlu dilakukan pemeriksaan kemungkinan terjadi kebocoran trafo sehingga mencemari kondisi oli. Akan kita lihat trend pada bulan mendatang. Perlu persiapan dilakukan PURIFIKASI Kami tetap meyarankan untuk ESP 2 dan 3 Unit 3 dilakukan PURIFIKASI dengan temperatur sekitar 700 GANTI MINYAK pada trafo ESP 2 dan 3 Unit 1 dan ESP 2 dan 3 Unit 2 For Test 25 July 2019 Kondisi trafo masih seperti bulan yang lalu Kami tetap meyarankan untuk ESP 1 Unit 1 dan ESP 2 dan 3 Unit 3 dilakukan PURIFIKASI dengan temperatur sekitar 70 ⁰ GANTI MINYAK pada trafo ESP 3 Unit 1 dan ESP 2 dan 3 Unit 2	
	4	28/06/2018			
	4	24/07/2018			
	4	25/08/2018			
	4	25/09/2018			
	4	23/10/2018			
	4	22/11/2018			
	4	08/12/2018			
	4	21/05/2019			
	4	20/06/2019			
	4	25/07/2019			
	4	20/08/2019			
	4	17/09/2019			
	4	17/10/2019			
	4	14/11/2019			
2	5	21/05/2018	Trafo 1 ESP		1
	5	28/06/2018			
	5	24/07/2018			
	5	25/08/2018			
	5	25/09/2018			
	5	23/10/2018			
	5	22/11/2018			
	5	08/12/2018			
	5	21/05/2019			
	5	20/06/2019			
	5	25/07/2019			
	5	20/08/2019			
	5	17/09/2019			
	5	17/10/2019			
	5	14/11/2019			
3	6	21/05/2018	Trafo 2 ESP	1	
	6	28/06/2018			
	6	24/07/2018			
	6	25/08/2018			
	6	25/09/2018			
	6	23/10/2018			
	6	22/11/2018			
	6	08/12/2018			
	6	21/05/2019			
	6	20/06/2019			
	6	25/07/2019			
	6	20/08/2019			
	6	17/09/2019			
	6	17/10/2019			
	6	14/11/2019			
4	7	21/05/2018	Trafo 3 ESP	1	
	7	28/06/2018			
	7	24/07/2018			
	7	25/08/2018			
	7	25/09/2018			
	7	23/10/2018			
	7	22/11/2018			
	7	08/12/2018			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix																
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612							
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	C ₂ H ₆	
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	ethane		
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	< 65	
	7	21/05/2019						0.051		33	1.5		C		19	B								
	7	20/06/2019						0.051	29.6	30	1.5	0.8689	C		19	B	16	17,771	65,478	6	73	829	0	
	7	25/07/2019						0.056		48	1.5		C		15	C								
	7	20/08/2019						0.051		43	1.5		C		18	D								
	7	17/09/2019						0.056		48	1.5		C		18	D								
	7	17/10/2019						0.059		52	1.5		C		19	B								
	7	14/11/2019						0.062	29.3	48	1.5	0.8613	C		18	C	13	19,116	69,791	7	58	909	0	
5	8	21/05/2018	Trafo 4 ESP	1	10034	2010	1,143	Fujia	0.034		37	1.5		C		23	D							
	8	28/06/2018							0.037	27.5	55	1.5	0.8653	C		20	D	9	17,962	64,248	4	75	1,054	0
	8	24/07/2018							0.039		50	1.5		C		19	D							
	8	25/08/2018							0.017		61	0.5		C		14	C							
	8	25/09/2018							0.014		45	0.5		C		20	D							
	8	23/10/2018							0.014		52	0.5		C		21	B							
	8	22/11/2018							0.014		44	0.5		C		21	C							
	8	08/12/2018							0.014		47	0.5		C		20	B							
	8	21/05/2019							0.014		47	0.5		C		19	B							
	8	20/06/2019							0.014	38.6	45	0.5	0.8775	C		19	B	11	15,056	64,828	5	11	951	0
	8	25/07/2019							0.014		67	0.5		C		16	C							
	8	20/08/2019							0.017		45	0.5		C		16	B							
	8	17/09/2019							0.017		52	0.5		C		16	B							
	8	17/10/2019							0.020		56	0.5		C		18	D							
	8	14/11/2019							0.020	38.8	58	0.5	0.8699	C		17	C	19	19,904	66,458	6	17	1,109	0
6	9	21/05/2018	Main trafo	3	1122170	2010	26,706	Shandong	0.014		58	0.5		C		11	B							
	9	28/06/2018							0.014	41.0	85	0.5	0.8805	C		8	A	18	11,431	48,283	13	700	1,512	1
	9	24/07/2018							0.014		77	0.5		C		8	A							
	9	25/08/2018							0.017		82	0.5		C		7	A							
	9	25/09/2018							0.017		83	0.5		C		6	A							
	9	23/10/2018							0.017		78	0.5		C		8	A							
	9	22/11/2018							0.014		63	0.5		C		7	A							
	9	08/12/2018							0.014		83	0.5		C		7	A							
	23	21/05/2019		3011160192	2018	30,183	PAUWES TRAF	0.011		93	0.5		C		7	A								
	23	20/06/2019							0.011	40.6	83	0.5	0.8786	C		8	A	3	9,903	31,346	6	156	369	0
	23	25/07/2019							0.014		87	0.5		C		6	A							
	23	20/08/2019							0.011		78	0.5		C		6	A							
	23	17/09/2019							0.014		81	0.5		C		6	A							
	23	17/10/2019							0.017		85	0.5		C		7	A							
	23	14/11/2019							0.014	39.4	69	0.5	0.8709	C		7	A	13	9,649	34,449	8	170	610	0
7	10	21/05/2018	Trafo UAT	3	1035082	2010	4,749	Shandong	0.020		81	0.5		C		14	B							
	10	28/06/2018							0.022	37.5	79	0.5	0.8793	C		10	A	35	13,798	68,180	22	1,144	3,644	5
	10	24/07/2018							0.020		76	0.5		C		11	A							
	10	25/08/2018							0.022		78	0.5		C		8	A							
	10	25/09/2018							0.020		75	0.5		C		8	A							
	10	23/10/2018							0.020		69	0.5		C		8	A							
	10	22/11/2018							0.022		80	0.5		C		11	A							
	10	08/12/2018							0.020		61	0.5		C		8	A							
	22	21/05/2019		PX18-0082	2018	12,000	B & D	0.014		85	0.5		C		5	A								
	22	20/06/2019							0.014	37.0	89	0.5	0.8798	C		5	A	4	10,251	34,223	16	39	232	26
	22	25/07/2019							0.014		83	0.5		C		5	A							
	22	20/08/2019							0.014		78	0.5		C		4	A							
	22	17/09/2019							0.014		88	0.5		C		4	A							
	22	17/10/2019							0.014		90	0.5		C		3	A							
	22	14/11/2019							0.014	35.1	85	0.5	0.8721	C		6	A	5	11,577	37,124	16	59	338	28
8	11	21/05/2018	Trafo 1 ESP	3	007308428	2010	731	Zhejian	0.022		31	0.5		C		27	D							



no	Test Cer.	Date Test	nama unit					lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION
					C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
					ethylene	acetylene		ASTM D924				ASTM D3635			ASTM D1275B		
			Standa		<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0.00	0.00	0.00		Non corrosive	
	7	21/05/2019															Perform OIL REGENERATION
	7	20/06/2019			1	0	96	0.000	0.520	0.00	57	0.01	0.00	0.02	2E	Non corrosive	Perform OIL REGENERATION
	7	25/07/2019															Perform OIL REGENERATION
	7	20/08/2019															Perform OIL REGENERATION
	7	17/09/2019															Perform OIL REGENERATION
	7	17/10/2019															Perform OIL REGENERATION
	7	14/11/2019			1	0	79	0.000	0.550	0.00	56	0.00	0.00	0.06	3A	Non corrosive	Perform OIL REGENERATION
5	8	21/05/2018	Trafo 4 ESP	1													Perform OIL REGENERATION
	8	28/06/2018			2	0	90	0.060	2.850	0.02	21	0.00	0.00	0.05	3 A	Suspec Corrosive	Perform OIL REGENERATION
	8	24/07/2018															Perform OIL REGENERATION
	8	25/08/2018															Periodic condition monitoring (change oil on beginning of August 2014)
	8	25/09/2018															Perform OIL PURIFICATION
	8	23/10/2018															Perform OIL PURIFIATION
	8	22/11/2018															Perform OIL PURIFIATION
	8	08/12/2018															Perform OIL PURIFIATION
	8	21/05/2019															Perform OIL PURIFIATION
	8	20/06/2019			0	0	27	0.010	1.120	0.00	0	0.00	0.00	0.02	3A	Suspec Corrosive	Periodic condition monitoring
	8	25/07/2019															Periodic condition monitoring
	8	20/08/2019															Periodic condition monitoring
	8	17/09/2019															Periodic condition monitoring
	8	17/10/2019															Periodic condition monitoring
	8	14/11/2019			0	0	42	0.010	1.090	0.00	0	0.00	0.00	0.03	3A	Non corrosive	Periodic condition monitoring
6	9	21/05/2018	Main trafo	3													Periodic condition monitoring
	9	28/06/2018			1	0	733	0.000	0.110	0.19	0	0.00	0.01	0.06	3 A	Suspec Corrosive	Add passivator
	9	24/07/2018															Periodic condition monitoring
	9	25/08/2018															Periodic condition monitoring
	9	25/09/2018															Periodic condition monitoring
	9	23/10/2018															Periodic condition monitoring
	9	22/11/2018															Periodic condition monitoring
	9	08/12/2018															Periodic condition monitoring
	23	21/05/2019															Periodic condition monitoring
	23	20/06/2019			0	0	165	0.000	0.000	0.00	0	0.00	0.00	0.02	3A	Non corrosive	Periodic condition monitoring
	23	25/07/2019															Periodic condition monitoring
	23	20/08/2019															Periodic condition monitoring
	23	17/09/2019															Periodic condition monitoring
	23	17/10/2019															Periodic condition monitoring
	23	14/11/2019			0	0	191	0.000	0.410	0.00	0	0.01	0.00	0.05	3A	Non corrosive	Periodic condition monitoring
7	10	21/05/2018	Trafo UAT	3													Periodic condition monitoring
	10	28/06/2018			7	1	1,214	0.000	0.140	0.21	0	0.00	0.01	0.11	3 A	Suspec Corrosive	Retest 6 months: DGA
	10	24/07/2018															Add Passivator
	10	25/08/2018															Periodic condition monitoring
	10	25/09/2018															Periodic condition monitoring
	10	23/10/2018															Periodic condition monitoring
	10	22/11/2018															Periodic condition monitoring
	10	08/12/2018															Periodic condition monitoring
	22	21/05/2019															Periodic condition monitoring
	22	20/06/2019			2	0	87	0.000	0.120	0.00	0	0.00	0.00	0.02			Periodic condition monitoring
	22	25/07/2019															Periodic condition monitoring
	22	20/08/2019															Periodic condition monitoring
	22	17/09/2019															Periodic condition monitoring
	22	17/10/2019															Periodic condition monitoring
	22	14/11/2019			2	0	110	0.000	0.190	0.00	10	0.00	0.00	0.05	3A	Non corrosive	Periodic condition monitoring
8	11	21/05/2018	Trafo 1 ESP	3													Perform OIL PURIFICATION



no	Test Cer.	Date Test	nama unit		PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar					
	7	21/05/2019			For Test 25 July 2019 Trend penurunan kondisi trafo ESP Jadi kami sarankan untuk ESP 1 Unit 1 dan ESP 2 dan 3 Unit 3 segera dilakukan PURIFIKASI dengan temperatur sekitar 70 ⁰ GANTI MINYAK pada trafo ESP 3 Unit 1 dan ESP 2 dan 3 Unit 2
	7	20/06/2019			
	7	25/07/2019			
	7	20/08/2019			
	7	17/09/2019			
	7	17/10/2019			
	7	14/11/2019			
5	8	21/05/2018	Trafo 4 ESP	1	
	8	28/06/2018			
	8	24/07/2018			
	8	25/08/2018			
	8	25/09/2018			
	8	23/10/2018			
	8	22/11/2018			
	8	08/12/2018			
	8	21/05/2019			
	8	20/06/2019			
	8	25/07/2019			
	8	20/08/2019			
	8	17/09/2019			
	8	17/10/2019			
	8	14/11/2019			
6	9	21/05/2018	Main trafo	3	
	9	28/06/2018			
	9	24/07/2018			
	9	25/08/2018			
	9	25/09/2018			
	9	23/10/2018			
	9	22/11/2018			
	9	08/12/2018			
	23	21/05/2019			
	23	20/06/2019			
	23	25/07/2019			
	23	20/08/2019			
	23	17/09/2019			
	23	17/10/2019			
	23	14/11/2019			
7	10	21/05/2018	Trafo UAT	3	
	10	28/06/2018			
	10	24/07/2018			
	10	25/08/2018			
	10	25/09/2018			
	10	23/10/2018			
	10	22/11/2018			
	10	08/12/2018			
	22	21/05/2019			
	22	20/06/2019			
	22	25/07/2019			
	22	20/08/2019			
	22	17/09/2019			
	22	17/10/2019			
	22	14/11/2019			
8	11	21/05/2018	Trafo 1 ESP	3	



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix																							
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612														
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	C ₂ H ₆								
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	ethane									
								BATASAN OPERASI								< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	< 65
	11	28/06/2018						0.022	35.0	60	0.5	0.8154	C		10	B	19	15,552	64,448	3	57	1,391	0								
	11	24/07/2018						0.020		53	0.5		C		9	B															
	11	25/08/2018						0.022		77	0.5		C		9	B															
	11	25/09/2018						0.020		64	0.5		C		10	B															
	11	23/10/2018						0.020		63	0.5		C		12	B															
	11	22/11/2018						0.020		53	0.5		C		9	B															
	11	08/12/2018						0.020		62	0.5		C		9	B															
	11	21/05/2019						0.022		85	0.5		C		9	B															
	11	20/06/2019						0.022	35.2	65	0.5	0.8212	C		9	B	11	19,448	64,974	6	15	1,064	0								
	11	25/07/2019						0.020		51	0.5		C		9	B															
	11	20/08/2019						0.020		69	0.5		C		8	B															
	11	17/09/2019						0.022		67	0.5		C		8	B															
	11	17/10/2019						0.022		76	0.5		C		8	A															
	11	14/11/2019						0.020	35.7	78	0.5	0.8140	C		8	A	16	20,510	67,519	6	31	1,063	0								
9	12	21/05/2018	Trafo 2 ESP	3	007008428	2010	731 Zhejian	0.022		28	0.5		C		30	E															
	12	28/06/2018						0.025	30.2	34	0.5	0.8080	C		16	C	43	22,272	68,719	4	38	1,226	0								
	12	24/07/2018						0.022		30	0.5		C		25	D															
	12	25/08/2018						0.022		67	0.5		C		11	C															
	12	25/09/2018						0.022		31	0.5		C		32	D															
	12	23/10/2018						0.025		54	0.5		C		13	B															
	12	22/11/2018						0.025		33	0.5		C		38	D															
	12	08/12/2018						0.025		30	0.5		C		32	C															
	12	21/05/2019						0.028		40	1.0		C		18	C															
	12	20/06/2019						0.028	31.3	41	1.0	0.8163	C		18	D	13	21,221	65,168	7	48	1,870	0								
	12	25/07/2019						0.031		60	1.0		C		21	D															
	12	20/08/2019						0.028		48	1.0		C		19	D															
	12	17/09/2019						0.031		43	1.0		C		20	D															
	12	17/10/2019						0.031		45	1.0		C		19	C															
	12	14/11/2019						0.028	31.1	48	1.0	0.8093	C		24	C	11	23,917	69,704	6	48	1,925	0								
10	13	21/05/2018	Trafo 3 ESP	3	007208428	2010	731 Zhejian	0.014		54	0.5		C		23	D															
	13	28/06/2018						0.017	38.4	43	0.5	0.8075	C		22	C	30	15,202	64,188	4	65	1,833	0								
	13	24/07/2018						0.014		43	0.5		C		20	C															
	13	25/08/2018						0.014		76	0.5		C		11	C															
	13	25/09/2018						0.020		47	0.5		C		20	B															
	13	23/10/2018						0.014		49	0.5		C		15	A															
	13	22/11/2018						0.017		39	0.5		C		15	A															
	13	08/12/2018						0.014		34	0.5		C		18	A															
	13	21/05/2019						0.014		41	0.5		C		20	C															
	13	20/06/2019						0.020	38.5	36	0.5	0.8215	C		23	C	14	18,302	64,811	6	86	2,181	0								
	13	25/07/2019						0.020		36	0.5		C		20	D															
	13	20/08/2019						0.017		32	0.5		C		19	C															
	13	17/09/2019						0.020		43	0.5		C		20	C															
	13	17/10/2019						0.017		36	0.5		C		21	B															
	13	14/11/2019						0.017	37.0	34	0.5	0.8143	C		26	B	10	16,999	64,949	8	91	1,921	0								
11	14	21/05/2018	Trafo 4 ESP	3	007108428	2010	731 Zhejian	0.028		37	1.0		C		21	D															
	14	28/06/2018						0.025	30.5	71	1.0	0.8442	C		11	A	20	14,690	67,822	2	22	683	0								
	14	24/07/2018						0.022		42	1.0		C		11	C															
	14	25/08/2018						0.025		66	1.5		C		11	B															
	14	25/09/2018						0.022		51	1.5		C		11	B															



no	Test Cer.	Date Test	nama unit					lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION
					C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
					ethylene	acethylene		ASTM D924				ASTM D3635			ASTM D1275B		
			Standa		<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0.00	0.00	0.00		Non corrosive	
	11	28/06/2018			0	0	79	0.000	0.050	0.16	0	0.00	0.00	0.02	1 B	Non corrosive	Retest 12 months: Specific Gravity Add Passivator
	11	24/07/2018															Periodic condition monitoring
	11	25/08/2018															Periodic condition monitoring (Purifying on end of July 2018)
	11	25/09/2018															Periodic condition monitoring
	11	23/10/2018															Periodic condition monitoring
	11	22/11/2018															Periodic condition monitoring
	11	08/12/2018															Periodic condition monitoring
	11	21/05/2019															Periodic condition monitoring
	11	20/06/2019			0	0	32	0.000	0.060	0.16	0	0.00	0.00	0.02	2E	Non corrosive	Periodic condition monitoring
	11	25/07/2019															Periodic condition monitoring
	11	20/08/2019															Periodic condition monitoring
	11	17/09/2019															Periodic condition monitoring
	11	17/10/2019															Periodic condition monitoring
	11	14/11/2019			0	0	53	0.000	0.240	0.16	0	0.00	0.00	0.03	2E	Non corrosive	Periodic condition monitoring
9	12	21/05/2018	Trafo 2 ESP	3													Perform OIL PURIFICATION
	12	28/06/2018			0	0	85	0.000	0.040	0.19	0	0.00	0.00	0.02	3 A	Suspec Corrosive	Retest 12 months: Specific Gravity Add Passivator
	12	24/07/2018															Perform OIL REGENERATION
	12	25/08/2018															Periodic condition monitoring (Purifying on end of July 2018)
	12	25/09/2018															Perform OIL REGENERATION
	12	23/10/2018															Periodic condition monitoring
	12	22/11/2018															Perform for Purification
	12	08/12/2018															Perform for Purification
	12	21/05/2019															Periodic condition monitoring
	12	20/06/2019			0	0	68	0.000	0.010	0.19	180	0.00	0.00	0.02	3A	Non corrosive	Periodic condition monitoring
	12	25/07/2019															Preparing for Purification
	12	20/08/2019															Preparing for Purification
	12	17/09/2019															Preparing for Purification
	12	17/10/2019															Preparing for Purification
	12	14/11/2019			0	0	65	0.000	0.010	0.19	181	0.00	0.00	0.01	3A	Non corrosive	Preparing for Purification
10	13	21/05/2018	Trafo 3 ESP	3													Perform OIL PURIFICATION
	13	28/06/2018			0	0	99	0.000	0.010	0.20	4	0.00	0.00	0.02	2 E	Non corrosive	Retest 12 months: Specific Gravity Retest 6 months: Dielectric and Water Content Add Passivator
	13	24/07/2018															Perform OIL PURIFICATION
	13	25/08/2018															Periodic condition monitoring (Purifying on end of July 2018)
	13	25/09/2018															Perform OIL PURIFICATION
	13	23/10/2018															Periodic condition monitoring
	13	22/11/2018															Periodic condition monitoring
	13	08/12/2018															Periodic condition monitoring
	13	21/05/2019															ReTest in 6 months : Water Content and BDV - Prepare for PURIFICATION
	13	20/06/2019			0	0	106	0.000	0.070	0.19	0	0.00	0.00	0.02	2E	Non corrosive	ReTest in 6 months : Water Content and BDV - Prepare for PURIFICATION
	13	25/07/2019															Preparing for Purification
	13	20/08/2019															Preparing for Purification
	13	17/09/2019															Preparing for Purification
	13	17/10/2019															Preparing for Purification
	13	14/11/2019			0	0	109	0.000	0.110	0.18	0	0.00	0.00	0.02	2D	Non corrosive	Preparing for Purification
11	14	21/05/2018	Trafo 4 ESP	3													Retest 6 months: Dielectric and Water Content
	14	28/06/2018			0	0	44	0.000	0.550	0.00	0	0.00	0.03	0.02	1 B	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	14	24/07/2018															Periodic condition monitoring
	14	25/08/2018															Periodic condition monitoring (Purifying on end of July 2018)
	14	25/09/2018															Periodic condition monitoring



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	11	28/06/2018			GANTI MINYAK Pada trafo ESP 3 Unit 1, ESP 2 dan 3 Unit 2 perlu dilakukan GANTI MINYAK, mengingat tingkat Acidity nya yang sudah tinggi.
	11	24/07/2018			
	11	25/08/2018			
	11	25/09/2018			
	11	23/10/2018			
	11	22/11/2018			
	11	08/12/2018			
	11	21/05/2019			
	11	20/06/2019			
	11	25/07/2019			
	11	20/08/2019			
	11	17/09/2019			
	11	17/10/2019			
	11	14/11/2019			
9	12	21/05/2018	Trafo 2 ESP	3	
	12	28/06/2018			
	12	24/07/2018			
	12	25/08/2018			
	12	25/09/2018			
	12	23/10/2018			
	12	22/11/2018			
	12	08/12/2018			
	12	21/05/2019			
	12	20/06/2019			
	12	25/07/2019			
	12	20/08/2019			
	12	17/09/2019			
	12	17/10/2019			
	12	14/11/2019			
10	13	21/05/2018	Trafo 3 ESP	3	
	13	28/06/2018			
	13	24/07/2018			
	13	25/08/2018			
	13	25/09/2018			
	13	23/10/2018			
	13	22/11/2018			
	13	08/12/2018			
	13	21/05/2019			
	13	20/06/2019			
	13	25/07/2019			
	13	20/08/2019			
	13	17/09/2019			
	13	17/10/2019			
	13	14/11/2019			
11	14	21/05/2018	Trafo 4 ESP	3	
	14	28/06/2018			
	14	24/07/2018			
	14	25/08/2018			
	14	25/09/2018			



no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	C ₂ H ₆
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	ethane	
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	< 65
	14	23/10/2018						0.022		56	1.5		C		12	B							
	14	22/11/2018						0.022		44	1.5		C		11	B							
	14	08/12/2018						0.022		49	1.0		C		13	C							
	14	21/05/2019						0.022		58	1.0		C		11	C							
	14	20/06/2019						0.022	30.3	51	1.0	0.8451	C		12	C	19	16,312	66,417	4	37	732	0
	14	25/07/2019						0.020		75	1.0		C		11	C							
	14	20/08/2019						0.022		49	1.0		C		11	B							
	14	17/09/2019						0.025		51	1.5		C		12	C							
	14	17/10/2019						0.028		51	1.0		C		14	C							
	14	14/11/2019						0.025	31.3	49	1.0	0.8440	C		14	B	13	18,447	65,198	6	34	877	0
12	15	21/05/2018	Ash Handling	10820108	2011	1,023	Wolong	0.022		33	1.0		C		26	C							
	15	28/06/2018						0.025	37.5	42	1.0	0.8804	C		24	C	53	10,939	69,394	20	557	5,425	9
	15	24/07/2018						0.022		39	1.0		C		20	B							
	15	25/08/2018						0.025		60	1.5		C		16	B							
	15	25/09/2018						0.025		43	1.5		C		17	A							
	15	23/10/2018						0.022		28	1.5		C		19	C							
	15	22/11/2018						0.022		24	1.5		C		28	B							
	15	08/12/2018						0.022		20	1.5		C		28	B							
	15	21/05/2019						0.022		28	1.5		C		29	C							
	15	20/06/2019						0.022	36.6	27	1.5	0.8890	C		29	C	53	14,032	65,585	13	487	5,360	0
	15	25/07/2019						0.025		49	1.0		C		21	B							
	15	20/08/2019						0.022		30	1.0		C		20	B							
	15	17/09/2019						0.025		52	1.5		C		21	B							
	15	17/10/2019						0.025		57	1.5		C		22	B							
	15	14/11/2019						0.028	38.2	47	1.0	0.8811	C		25	C	42	13,399	67,123	22	661	6,026	11
13	16	21/05/2018	Ash Handling	10820107	2011	1,023	Wolong	0.020		24	1.5		C		32	D							
	16	28/06/2018						0.022	35.0	50	1.5	0.8811	C		16	B	32	22,904	67,139	4	249	3,112	0
	16	24/07/2018						0.020		27	1.5		C		30	C							
	16	25/08/2018						0.022		45	1.5		C		16	B							
	16	25/09/2018						0.022		36	1.5		C		19	B							
	16	23/10/2018						0.022		36	1.5		C		23	B							
	16	22/11/2018						0.022		30	1.5		C		25	B							
	16	08/12/2018						0.022		32	1.5		C		24	B							
	16	21/05/2019						0.022		32	1.5		C		23	B							
	16	20/06/2019						0.022	36.8	37	1.5	0.8889	C		24	B	39	11,205	66,397	21	439	5,998	11
	16	25/07/2019						0.025		42	1.5		C		27	B							
	16	20/08/2019						0.025		36	1.5		C		29	B							
	16	17/09/2019						0.028		37	1.5		C		26	C							
	16	17/10/2019						0.025		37	1.5		C		27	C							
	16	14/11/2019						0.022	34.6	29	1.5	0.8809	C		31	D	18	16,725	64,652	7	296	4,058	0
14	17	21/05/2018	Trafo 1 ESP	2	10037	1010	731	Fujian		0.025	24	1.0		C		32	E						
	17	28/06/2018							32.9	77	1.0	0.8693	C		11	B	93	17,225	43,231	5	57	775	0
	17	24/07/2018						0.028		69	1.0		C		11	B							
	17	25/08/2018						0.031		74	1.5		C		12	B							
	17	25/09/2018						0.028		74	1.5		C		10	B							
	17	23/10/2018						0.028		54	1.5		C		11	B							
	17	22/11/2018						0.028		54	1.5		C		10	B							
	17	08/12/2018						0.028		46	1.5		C		10	B							
	17	21/05/2019						0.031		67	1.5		C		12	B							
	17	20/06/2019						0.031	31.0	62	1.0	0.8776	C		13	C	62	18,560	46,166	5	50	839	0



no	Test Cer.	Date Test	nama unit												MALAYSIAN LAB RECOMMENDATION		
							lpf (C)		dbpc	furan	metal in oil			corr. sul		keterangan	
				C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu				
Standa				ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B			
					<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0.00	0.00	0.00		Non corrosive	
	14	23/10/2018															Periodic condition monitoring
	14	22/11/2018															Periodic condition monitoring
	14	08/12/2018															Periodic condition monitoring
	14	21/05/2019															Periodic condition monitoring
	14	20/06/2019			0	0	60	0.000	0.490	0.00	0	0.00	0.03	0.01	1B	Non corrosive	Periodic condition monitoring
	14	25/07/2019															Periodic condition monitoring
	14	20/08/2019															Periodic condition monitoring
	14	17/09/2019															Periodic condition monitoring
	14	17/10/2019															Periodic condition monitoring
	14	14/11/2019			0	0	53	0.000	0.340	0.00	0	0.00	0.00	0.03	2E	Non corrosive	Periodic condition monitoring
12	15	21/05/2018	Ash Handling														Periodic condition monitoring
	15	28/06/2018			8	0	647	0.000	0.460	0.22	0	0.00	0.01	0.04	1 B	Non corrosive	Add passivator
	15	24/07/2018															Periodic condition monitoring
	15	25/08/2018															Periodic condition monitoring
	15	25/09/2018															Periodic condition monitoring
	15	23/10/2018															Periodic condition monitoring
	15	22/11/2018															Monitoring : Water Content and BDV - Prepare for PURIFICATION
	15	08/12/2018															Retest in 6 months : Water Content and BDV - Prepare for PURIFICATION
	15	21/05/2019															Perform for PURIFICATION
	15	20/06/2019			11	0	564	0.000	0.360	0.21	0	0.00	0.00	0.02	2e	Non corrosive	Perform for PURIFICATION
	15	25/07/2019															Periodic condition monitoring
	15	20/08/2019															Periodic condition monitoring
	15	17/09/2019															Periodic condition monitoring
	15	17/10/2019															Periodic condition monitoring
	15	14/11/2019			14	ND	750	0.000	0.230	0.21	21	0.00	0.00	0.04	3A	Non corrosive	Periodic condition monitoring; Check Cooling system Prepare for Purification
13	16	21/05/2018	Ash Handling														Perform OIL PURIFICATION
	16	28/06/2018			5	0	290	0.000	0.490	0.20	0	0.00	0.00	0.02	3 A	Suspec Corrosive	Retest 6 months: DGA Add Passivator
	16	24/07/2018															Perform OIL PURIFICATION
	16	25/08/2018															Periodic condition monitoring (Purifying on end of July 2018)
	16	25/09/2018															Periodic condition monitoring
	16	23/10/2018															Periodic condition monitoring
	16	22/11/2018															Periodic condition monitoring
	16	08/12/2018															Periodic condition monitoring
	16	21/05/2019															Periodic condition monitoring
	16	20/06/2019			9	0	519	0.000	0.680	0.20	0	0.01	0.00	0.02	3A	Non corrosive	Periodic condition monitoring
	16	25/07/2019															Periodic condition monitoring
	16	20/08/2019															Periodic condition monitoring
	16	17/09/2019															Periodic condition monitoring
	16	17/10/2019															Periodic condition monitoring
	16	14/11/2019			6	0	329	0.000	0.550	0.19	0	0.00	0.00	0.06	3A	Non corrosive	Periodic condition monitoring; Check Cooling system Prepare for Purification
14	17	21/05/2018	Trafo 1 ESP	2													OIL PURIFICATION
	17	28/06/2018			2	0	157	0.030	3.420	0.00	0	0.00	0.06	0.03	2 C	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	17	24/07/2018															Periodic condition monitoring
	17	25/08/2018															Periodic condition monitoring
	17	25/09/2018															Periodic condition monitoring
	17	23/10/2018															Periodic condition monitoring
	17	22/11/2018															Periodic condition monitoring
	17	08/12/2018															Periodic condition monitoring
	17	21/05/2019															Periodic condition monitoring
	17	20/06/2019			0	0	117	0.000	0.720	0.00	0	0.00	0.00	0.02	3A	Non corrosive	Monitoring : Acidity, IFT and LPF



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	14	23/10/2018			
	14	22/11/2018			
	14	08/12/2018			
	14	21/05/2019			
	14	20/06/2019			
	14	25/07/2019			
	14	20/08/2019			
	14	17/09/2019			
	14	17/10/2019			
	14	14/11/2019			
12	15	21/05/2018	Ash Handling		
	15	28/06/2018			
	15	24/07/2018			
	15	25/08/2018			
	15	25/09/2018			
	15	23/10/2018			
	15	22/11/2018			
	15	08/12/2018			
	15	21/05/2019			
	15	20/06/2019			
	15	25/07/2019			
	15	20/08/2019			
	15	17/09/2019			
	15	17/10/2019			
	15	14/11/2019			
13	16	21/05/2018	Ash Handling		
	16	28/06/2018			
	16	24/07/2018			
	16	25/08/2018			
	16	25/09/2018			
	16	23/10/2018			
	16	22/11/2018			
	16	08/12/2018			
	16	21/05/2019			
	16	20/06/2019			
	16	25/07/2019			
	16	20/08/2019			
	16	17/09/2019			
	16	17/10/2019			
	16	14/11/2019			
14	17	21/05/2018	Trafo 1 ESP	2	
	17	28/06/2018			
	17	24/07/2018			
	17	25/08/2018			
	17	25/09/2018			
	17	23/10/2018			
	17	22/11/2018			
	17	08/12/2018			
	17	21/05/2019			
	17	20/06/2019			



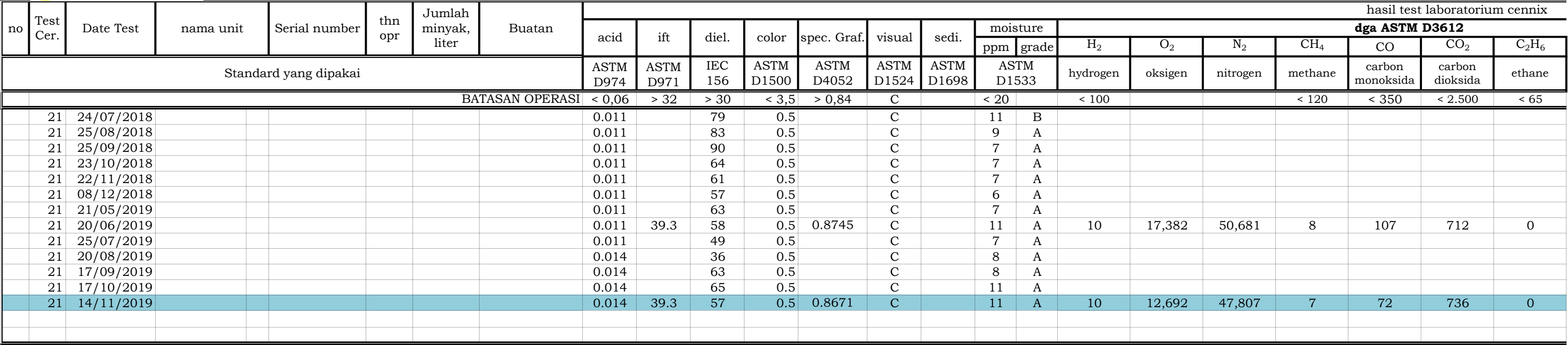
no	Test Cer.	Date Test	nama unit	Serial number	thn opr	Jumlah minyak, liter	Buatan	hasil test laboratorium cennix															
								acid	ift	diel.	color	spec. Graf.	visual	sedi.	moisture		dga ASTM D3612						
															ppm	grade	H ₂	O ₂	N ₂	CH ₄	CO	CO ₂	C ₂ H ₆
Standard yang dipakai								ASTM D974	ASTM D971	IEC 156	ASTM D1500	ASTM D4052	ASTM D1524	ASTM D1698	ASTM D1533	hydrogen	oksigen	nitrogen	methane	carbon monoksida	carbon dioksida	ethane	
		BATASAN OPERASI						< 0,06	> 32	> 30	< 3,5	> 0,84	C		< 20		< 100			< 120	< 350	< 2.500	< 65
	17	25/07/2019						0.031		64	1.0		C		11	B							
	17	20/08/2019						0.034		63	1.0		C		11	B							
	17	17/09/2019						0.034		19	1.0		C		62	E							
	17	17/10/2019						0.037		91	1.0		C		16	B							
	17	14/11/2019			1010		731	0.031	30.7	41	1.0	0.8700	C		15	B	17	18,543	51,095	6	35	992	0
15	18	21/05/2018	Trafo 2 ESP	2	10036			0.053		55	1.0		C		16	C							
	18	28/06/2018						0.056	31.3	66	1.0	0.8573	C		13	C	8	22,714	64,906	5	67	879	0
	18	24/07/2018						0.059		79	1.5		C		8	A							
	18	25/08/2018						0.062		67	1.5		C		10	B							
	18	25/09/2018						0.056		51	1.5		C		9	B							
	18	23/10/2018						0.062		67	1.5		C		13	B							
	18	22/11/2018						0.062		61	1.5		C		8	A							
	18	08/12/2018						0.062		59	1.5		C		8	A							
	18	21/05/2019						0.065		77	1.0		C		11	B							
	18	20/06/2019						0.065	29.7	76	1.0	0.8655	C		12	B	5	29,439	65,338	5	25	977	0
	18	25/07/2019						0.062		47	1.5		C		11	B							
	18	20/08/2019						0.065		41	1.5		C		12	B							
	18	17/09/2019						0.067		59	1.5		C		11	B							
	18	17/10/2019						0.065		74	1.5		C		11	B							
	18	14/11/2019						0.062	30.5	77	1.0	0.8586	C		12	B	5	28,110	64,911	6	43	1,069	0
16	19	21/05/2018	Trafo 3 ESP	2	10033		731	0.059		51	1.5		C		18	D							
	19	28/06/2018						0.062	26.3	78	1.5	0.8602	C		14	C	14	17,627	48,672	4	47	935	0
	19	24/07/2018						0.070		67	1.5		C		12	B							
	19	25/08/2018						0.067		73	1.5		C		13	C							
	19	25/09/2018						0.067		59	1.5		C		12	B							
	19	23/10/2018						0.070		86	1.5		C		13	B							
	19	22/11/2018						0.070		13	1.5		C		84	E							
	19	08/12/2018						0.070		8	1.5		C		84	E							
	19	21/05/2019						0.073		50	1.5		C		13	C							
	19	20/06/2019						0.073	26.5	53	1.5	0.8687	C		13	C	16	17,449	51,923	5	58	1,049	0
	19	25/07/2019						0.070		48	1.5		C		13	C							
	19	20/08/2019						0.067		58	1.5		C		14	B							
	19	17/09/2019						0.062		84	1.5		C		10	B							
	19	17/10/2019						0.067		77	1.5		C		12	B							
	19	14/11/2019						0.062	26.5	71	1.5		C		13	B	18	21,705	58,102	6	59	1,088	0
17	20	21/05/2018	Trafo 4 ESP	2	10032		731	0.025		40	1.0		C		18	D							
	20	28/06/2018						0.028	32.6	88	1.0	0.8648	C		15	D	22	25,201	74,464	5	85	823	0
	20	24/07/2018						0.031		59	1.0		C		13	B							
	20	25/08/2018						0.031		68	1.5		C		10	B							
	20	25/09/2018						0.031		51	1.5		C		11	B							
	20	23/10/2018						0.034		57	1.0		C		15	C							
	20	22/11/2018						0.034		69	1.0		C		14	B							
	20	08/12/2018						0.034		61	1.0		C		18	D							
	20	21/05/2019						0.036		60	1.0		C		14	B							
	20	20/06/2019						0.036	30.3	61	1.0	0.8732	C		14	C	33	25,208	70,145	4	35	853	0
	20	25/07/2019						0.034		59	1.0		C		12	B							
	20	20/08/2019						0.036		45	1.5		C		12	B							
	20	17/09/2019						0.034		50	1.5		C		12	B							
	20	17/10/2019						0.037		78	1.5		C		12	B							
	20	14/11/2019						0.034	30.2	50	1.0	0.8659	C		18	C	16	24,508	71,482	5	31	793	0
18	21	21/05/2018	Main Trafo	2	1006902		2011	0.011		46	0.5		C		14	B							
	21	28/06/2018						0.008	42.1	75	0.5	0.8663	C		10	B	19	12,733	53,386	10	193	2,448	0



no	Test Cer.	Date Test	nama unit					lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION
					C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
					ethylene	acetylene		ASTM D924				ASTM D3635			ASTM D1275B		
			Standa		<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0.00	0.00	0.00		Non corrosive	
	17	25/07/2019															Periodic condition monitoring
	17	20/08/2019															Periodic condition monitoring
	17	17/09/2019															Perform OIL PURIFICATION
	17	17/10/2019															Periodic condition monitoring
	17	14/11/2019			0	0	58	0.010	0.610	0.00	0	0.00	0.00	0.03	3A	Non corrosive	Periodic condition monitoring
15	18	21/05/2018	Trafo 2 ESP	2													Retest 6 months: Acidity and IFT
	18	28/06/2018			0	0	80	0.000	0.680	0.00	42	0.00	0.00	0.02	3 A	Suspec Corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	18	24/07/2018															Periodic condition monitoring
	18	25/08/2018															Periodic condition monitoring
	18	25/09/2018															Perform OIL REGENERATION
	18	23/10/2018															Perform OIL REGENERATION
	18	22/11/2018															Perform OIL REGENERATION
	18	08/12/2018															Perform OIL REGENERATION
	18	21/05/2019															Perform OIL REGENERATION
	18	20/06/2019			0	0	35	0.000	0.450	0.00	0	0.00	0.00	0.02	3A	Non corrosive	Perform OIL REGENERATION
	18	25/07/2019															Perform OIL REGENERATION
	18	20/08/2019															Perform OIL REGENERATION
	18	17/09/2019															Perform OIL REGENERATION
	18	17/10/2019															Perform OIL REGENERATION
	18	14/11/2019			0	0	54	0.000	0.640	0.00	0	0.00	0.00	0.03	2E	Non corrosive	Perform OIL REGENERATION
16	19	21/05/2018	Trafo 3 ESP	2													Retest 6 months: Acidity and IFT
	19	28/06/2018			0	0	65	0.000	0.570	0.00	0	0.00	0.00	0.02	3 A	Suspec Corrosive	Perform OIL REGENERATION
	19	24/07/2018															Perform OIL REGENERATION
	19	25/08/2018															Perform OIL REGENERATION
	19	25/09/2018															Perform OIL REGENERATION
	19	23/10/2018															Perform OIL REGENERATION
	19	22/11/2018															Perform OIL REGENERATION
	19	08/12/2018															Perform OIL REGENERATION
	19	21/05/2019															Perform OIL REGENERATION
	19	20/06/2019			0	0	79	0.010	0.790	0.00	0	0.00	0.00	0.02	3A	Non corrosive	Perform OIL REGENERATION
	19	25/07/2019															Perform OIL REGENERATION
	19	20/08/2019															Perform OIL REGENERATION
	19	17/09/2019															Perform OIL REGENERATION
	19	17/10/2019															Perform OIL REGENERATION
	19	14/11/2019			0	0	83	0.000	0.580	0.00	0	0.00	0.00	0.04	3A	Non corrosive	Perform OIL REGENERATION
17	20	21/05/2018	Trafo 4 ESP	2													Periodic condition monitoring
	20	28/06/2018			0	0	112	0.000	1.620	0.01	0	0.00	0.00	0.03	2 C	Non corrosive	Add inhibitor 2,6-DBPC and PASSIVATOR
	20	24/07/2018															Periodic condition monitoring
	20	25/08/2018															Periodic condition monitoring
	20	25/09/2018															Periodic condition monitoring
	20	23/10/2018															Periodic condition monitoring
	20	22/11/2018															Periodic condition monitoring
	20	08/12/2018															Periodic condition monitoring
	20	21/05/2019															Periodic condition monitoring
	20	20/06/2019			0	0	72	0.020	1.150	0.01	7	0.00	0.00	0.02	2C	Non corrosive	Monitoring : Acidity, IFT and LPF
	20	25/07/2019															Periodic condition monitoring
	20	20/08/2019															Periodic condition monitoring
	20	17/09/2019															Periodic condition monitoring
	20	17/10/2019															Periodic condition monitoring
	20	14/11/2019			0	0	52	0.000	0.960	0.04	0	0.00	0.00	0.03	3A	Non corrosive	Periodic condition monitoring
18	21	21/05/2018	Main Trafo	2													Periodic condition monitoring
	21	28/06/2018			2	0	224	0.000	0.410	0.22	113	0.00	0.00	0.03	1 B	Non corrosive	Retest 6 months: Furan and DGA Add Passivator



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION	
Standar					
	17	25/07/2019			
	17	20/08/2019			
	17	17/09/2019			
	17	17/10/2019			
	17	14/11/2019			
15	18	21/05/2018	Trafo 2 ESP	2	
	18	28/06/2018			
	18	24/07/2018			
	18	25/08/2018			
	18	25/09/2018			
	18	23/10/2018			
	18	22/11/2018			
	18	08/12/2018			
	18	21/05/2019			
	18	20/06/2019			
	18	25/07/2019			
	18	20/08/2019			
	18	17/09/2019			
	18	17/10/2019			
	18	14/11/2019			
16	19	21/05/2018	Trafo 3 ESP	2	
	19	28/06/2018			
	19	24/07/2018			
	19	25/08/2018			
	19	25/09/2018			
	19	23/10/2018			
	19	22/11/2018			
	19	08/12/2018			
	19	21/05/2019			
	19	20/06/2019			
	19	25/07/2019			
	19	20/08/2019			
	19	17/09/2019			
	19	17/10/2019			
	19	14/11/2019			
17	20	21/05/2018	Trafo 4 ESP	2	
	20	28/06/2018			
	20	24/07/2018			
	20	25/08/2018			
	20	25/09/2018			
	20	23/10/2018			
	20	22/11/2018			
	20	08/12/2018			
	20	21/05/2019			
	20	20/06/2019			
	20	25/07/2019			
	20	20/08/2019			
	20	17/09/2019			
	20	17/10/2019			
	20	14/11/2019			
18	21	21/05/2018	Main Trafo	2	
	21	28/06/2018			





no	Test Cer.	Date Test	nama unit					lpf (C)		dbpc	furan	metal in oil			corr. sul	keterangan	MALAYSIAN LAB RECOMMENDATION
					C ₂ H ₄	C ₂ H ₂	TOT	25 ^o	100 ^o			Al	Fe	Cu			
					ethylene	acethylene		ASTM D924		ASTM D2668	ASTM D5837	ASTM D3635			ASTM D1275B		
					<50	< 2	< 687	< 0.1	< 2.9	> .2	< 100	0.00	0.00	0.00		Non corrosive	
	21	24/07/2018															Periodic Condition monitoring
	21	25/08/2018															Periodic condition monitoring
	21	25/09/2018															Periodic condition monitoring
	21	23/10/2018															Periodic condition monitoring
	21	22/11/2018															Periodic condition monitoring
	21	08/12/2018															Periodic condition monitoring
	21	21/05/2019															Periodic condition monitoring
	21	20/06/2019			2	0	127	0.010	0.730	0.22	0	0.00	0.00	0.02	2E	Non corrosive	Periodic condition monitoring
	21	25/07/2019															Periodic condition monitoring
	21	20/08/2019															Periodic condition monitoring
	21	17/09/2019															Periodic condition monitoring
	21	17/10/2019															Periodic condition monitoring
	21	14/11/2019			2	0	91	0.020	0.600	0.22	0	0.00	0.00	0.03	3A	Non corrosive	Periodic condition monitoring

Summary by Soetjahjono
TxM Indonesia



no	Test Cer.	Date Test	nama unit	PAK SOETJAHJONO (TXM ADVISOR) RECOMMENDATION
Standar				
	21	24/07/2018		
	21	25/08/2018		
	21	25/09/2018		
	21	23/10/2018		
	21	22/11/2018		
	21	08/12/2018		
	21	21/05/2019		
	21	20/06/2019		
	21	25/07/2019		
	21	20/08/2019		
	21	17/09/2019		
	21	17/10/2019		
	21	14/11/2019		